

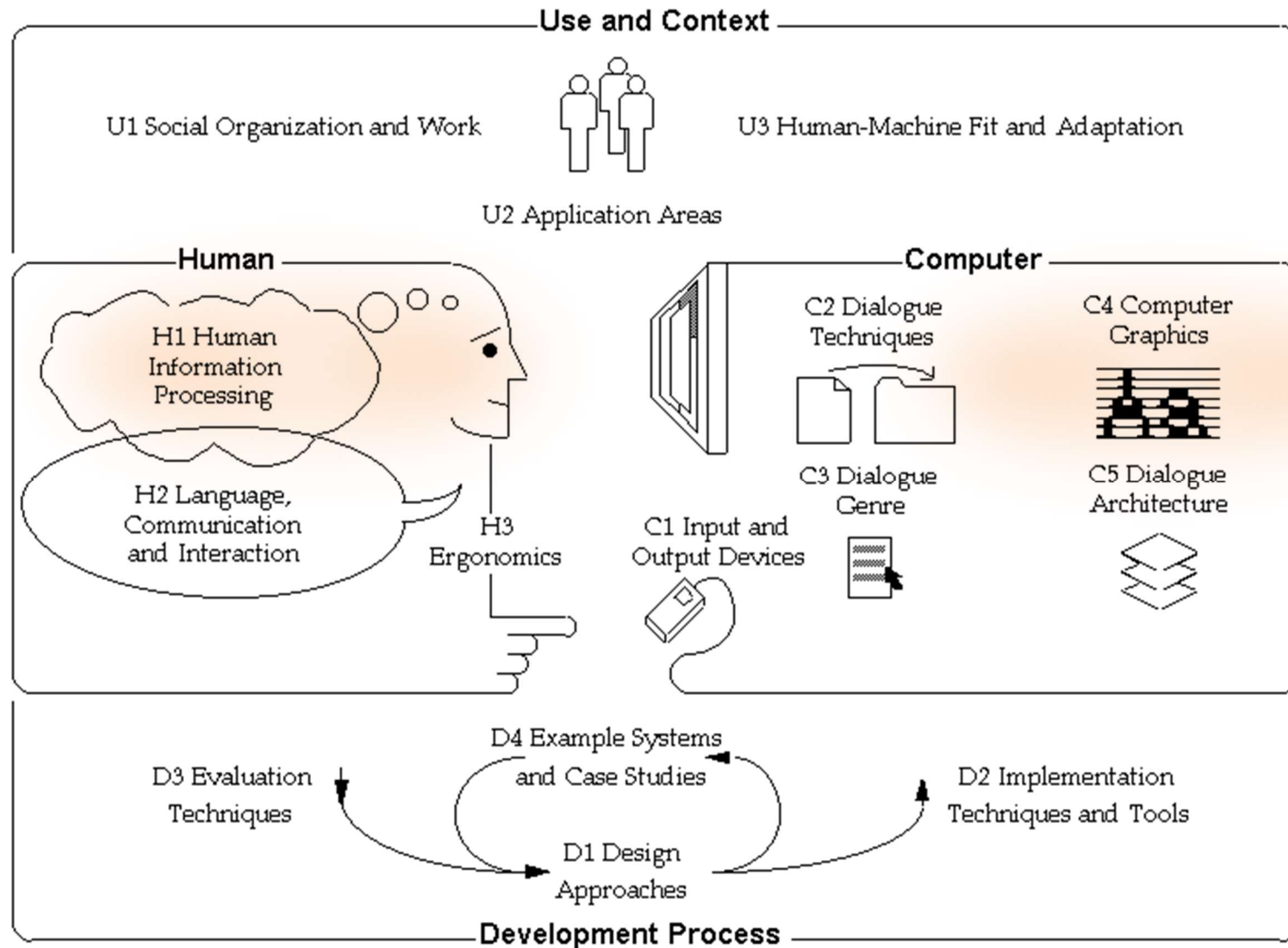
Human-Computer Interaction

Johan F. HOORN and Jeff TANG

Lecture 03, Part 1: Visual perception and computer graphics



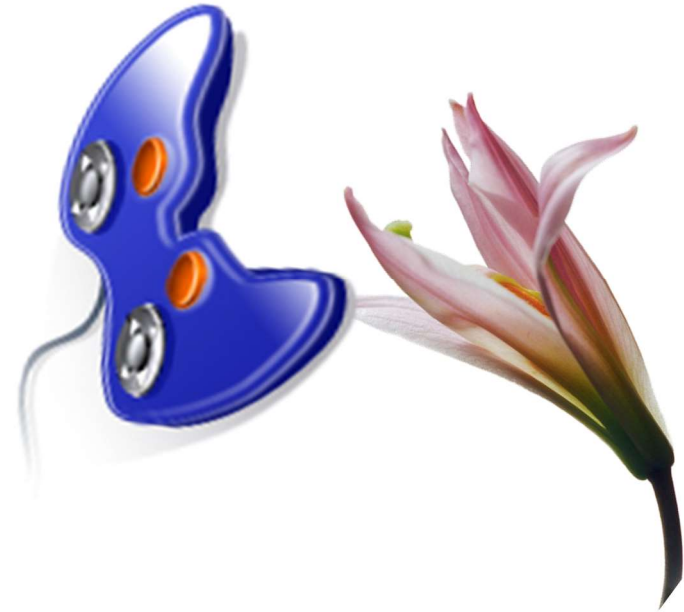
Overview



Contents

□ Visual

- Image
 - Physical reception
 - Interpretation: 3D, color
 - Images in design
- Text
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Text in design
- Computer Graphics

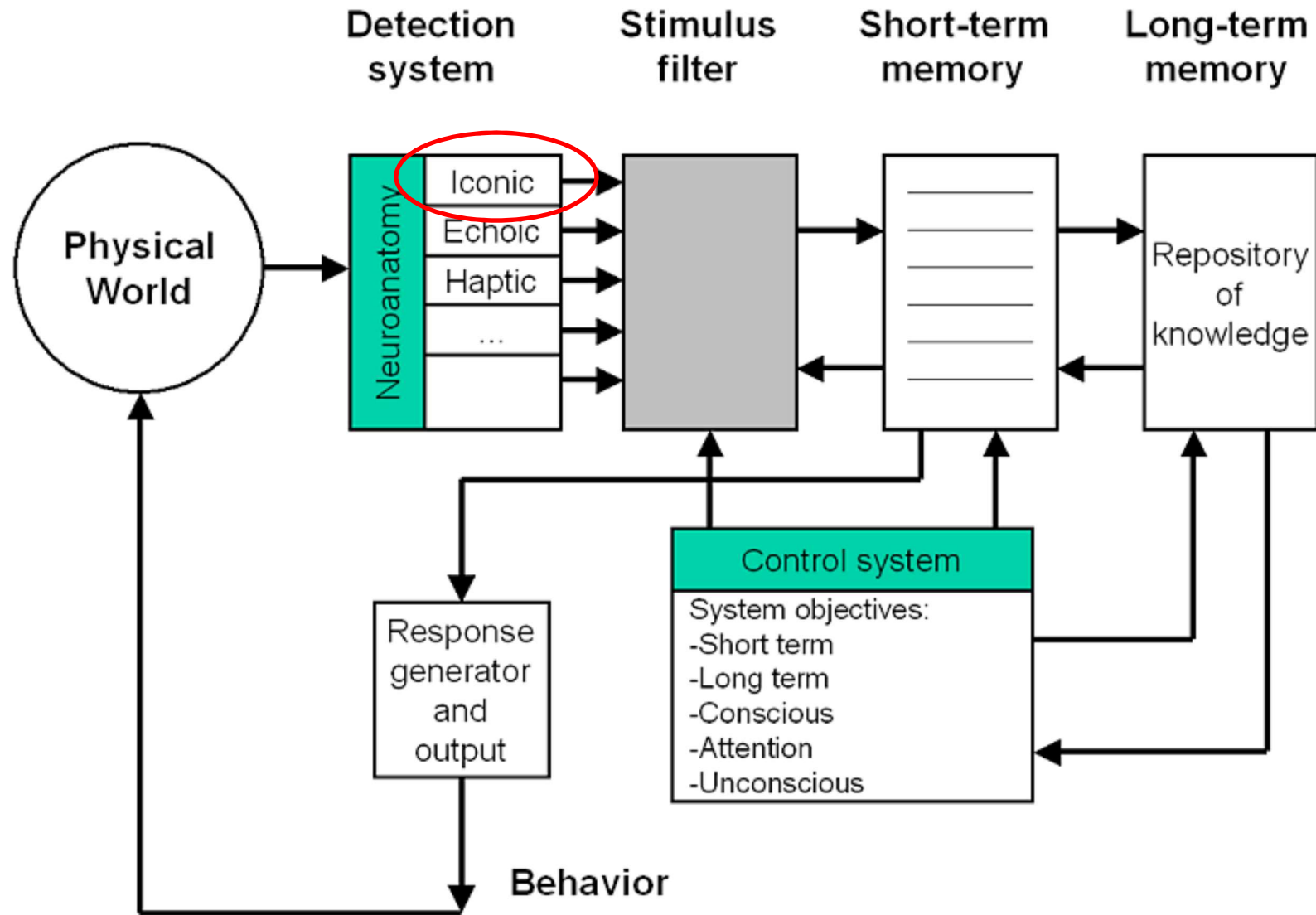


What you'll learn today

- ❑ Human visual system
- ❑ Perceiving images
- ❑ Detecting patterns
- ❑ Deciphering texts
- ❑ How subjective and deceptive all this is
- ❑ Consequences for design



The Human Information Processor

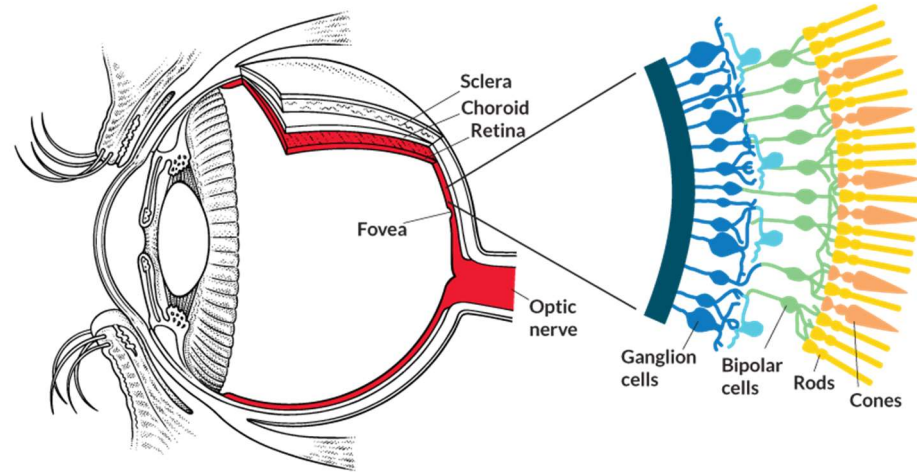


Visual system

Input: perception

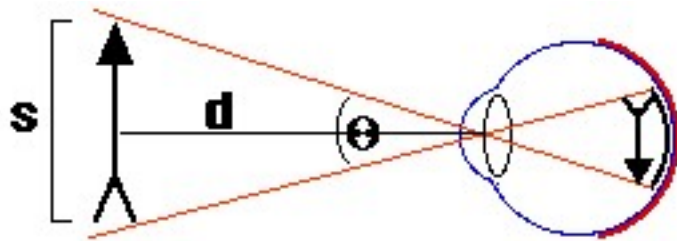
□ Visual ('iconic')

- Image
 - Physical reception



Eye receives light and transforms it into electrical energy

Light reflects from objects; their images are focused upside-down on retina



Law of visual angle

$$\tan \theta/2 = s/2d$$

$$\theta = 2 \arctan(s/2d)$$

$$\theta \quad s = 2d \tan \theta/2$$

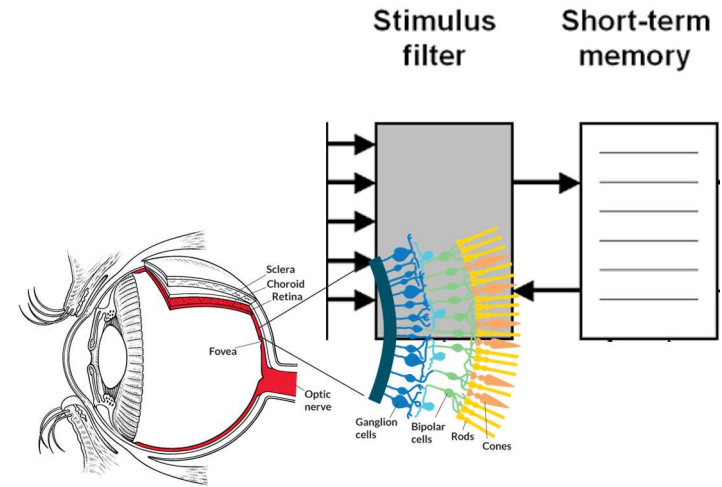
Ganglion cells detect pattern and movement

Retina contains rods for low-light vision and cones for color vision

Input: perception

❑ Visual ('iconic')

- Image
 - Aspects of physical reception



Brightness

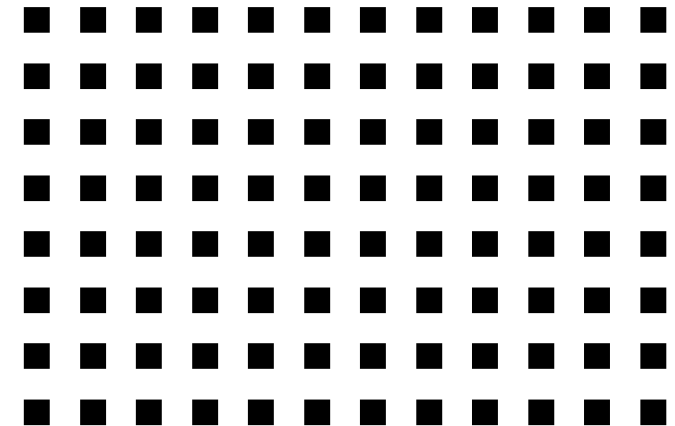
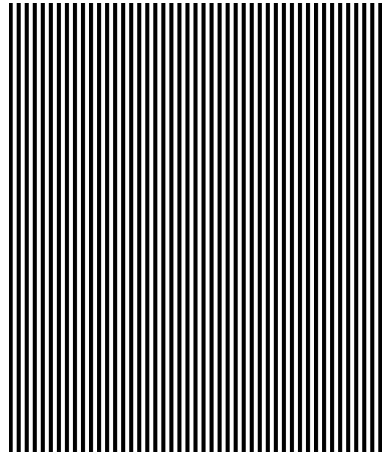
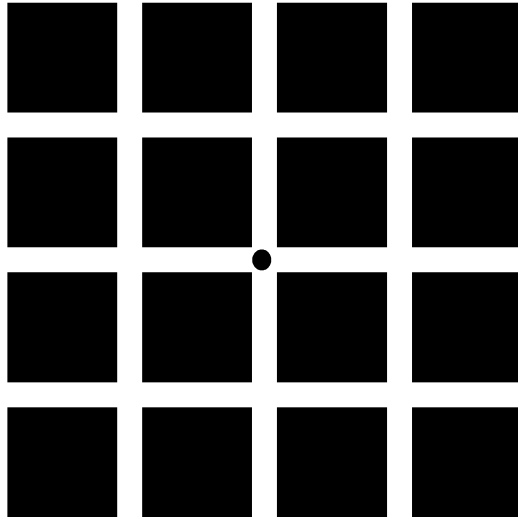
- subjective reaction to levels of light
- affected by luminance of object
- measured by just noticeable difference
- visual acuity increases with luminance as does flashing



Pattern filters (ganglion cells)

- certain neurons work as pattern filters
- filters pass middle frequencies better than higher and lower frequencies
- mutual suppression
(random noise from individual cells is corrected by the larger group)

Hermann grid



Springer lines

Are the grey spots and lines visual input or output?

Which color reflects the higher frequency?

Which the lower?

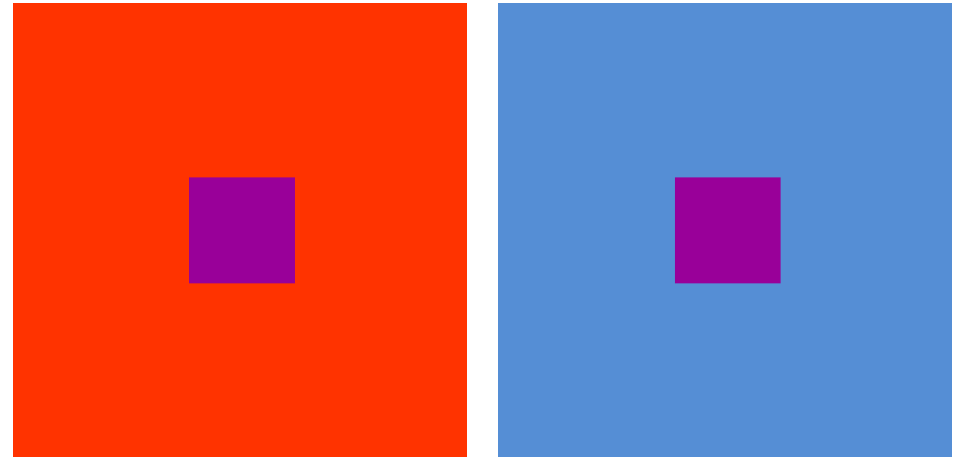
What the middle?

Is vision completely data-driven?

Input: perception

❑ Visual

- Image
 - Physical reception



Color

- Hue (based on the wavelength of light)
- Intensity (lighter/darker)
- Saturation (purity of the hue in a color)
- The higher the saturation, the purer the color
- Cones sensitive to color wavelengths
- Again, mutual suppression
- Blue acuity is lowest
- 8% males and 1% females are color blind

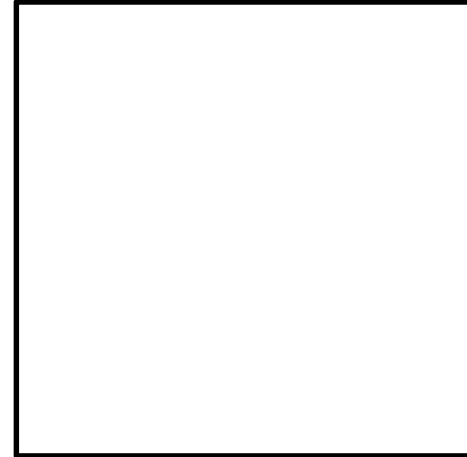
Input: perception

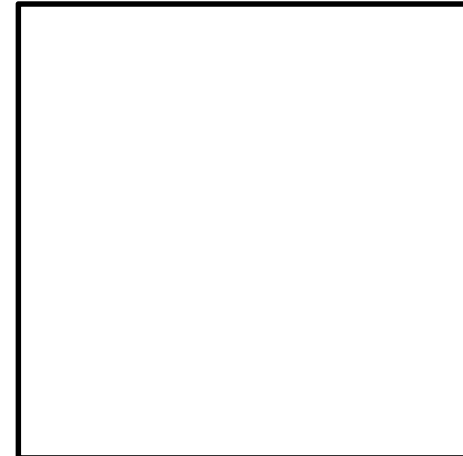
□ Visual

- Image
 - Physical reception

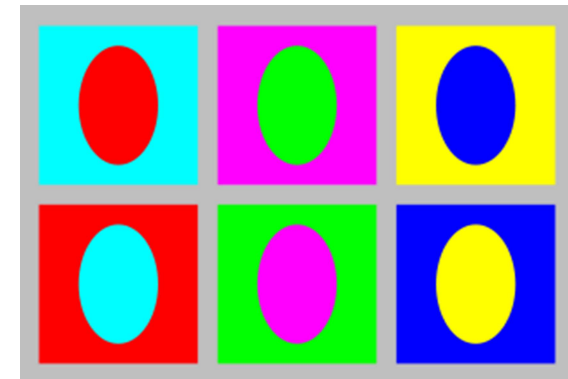


Which square is the darkest?





- Cones for red wear out
- Mutual suppression (e.g., red over green)
- Cones for cyan remain firing
- After-images



Complementary colors
in the [RGB color model](#)

Input: perception

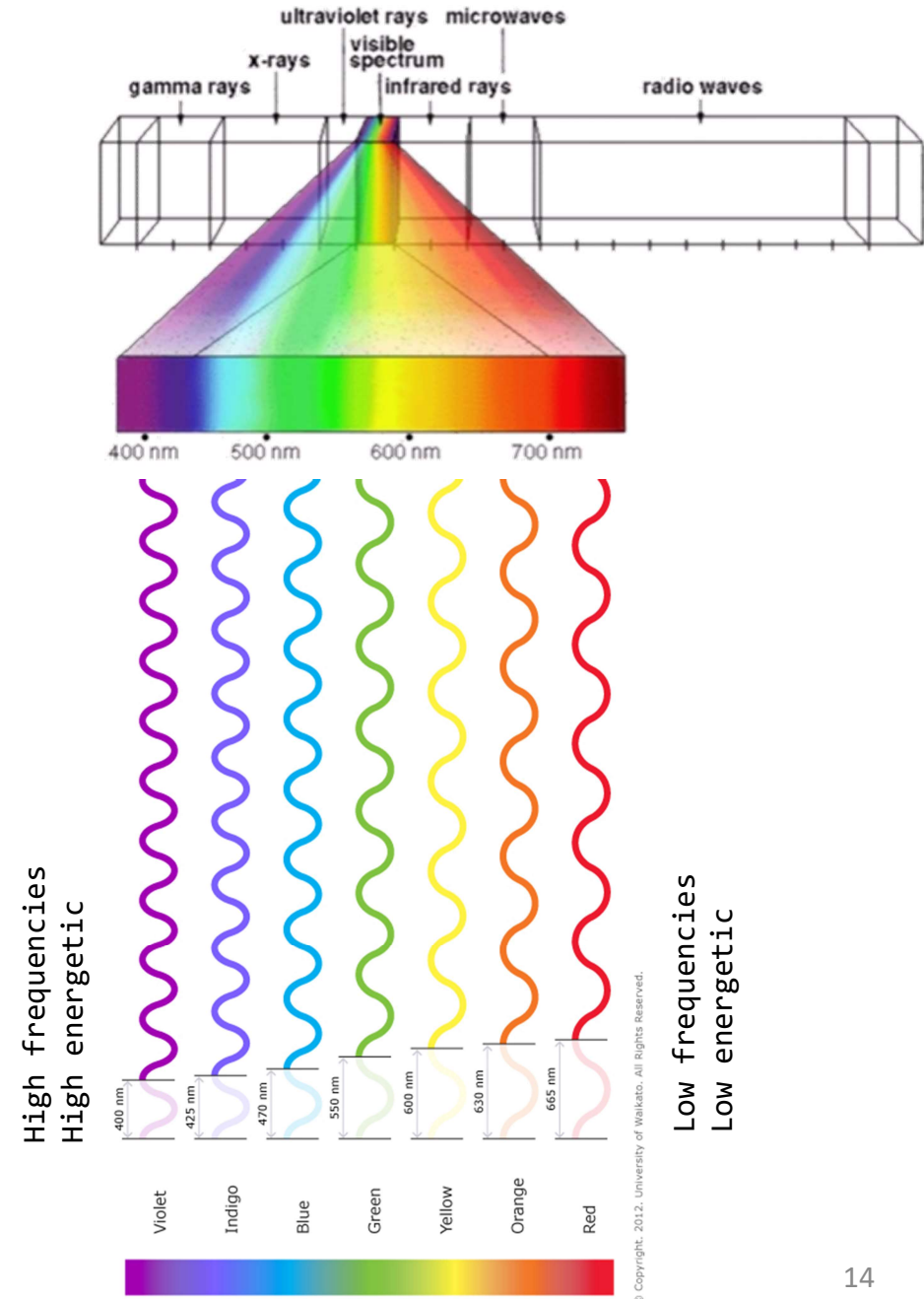
□ Visual

- Image
 - Interpretation: color

Note:

Color is **NO** property of the physics of light

Color is the human interpretation of different energetic frequencies (short waves, long waves) that hit a cell



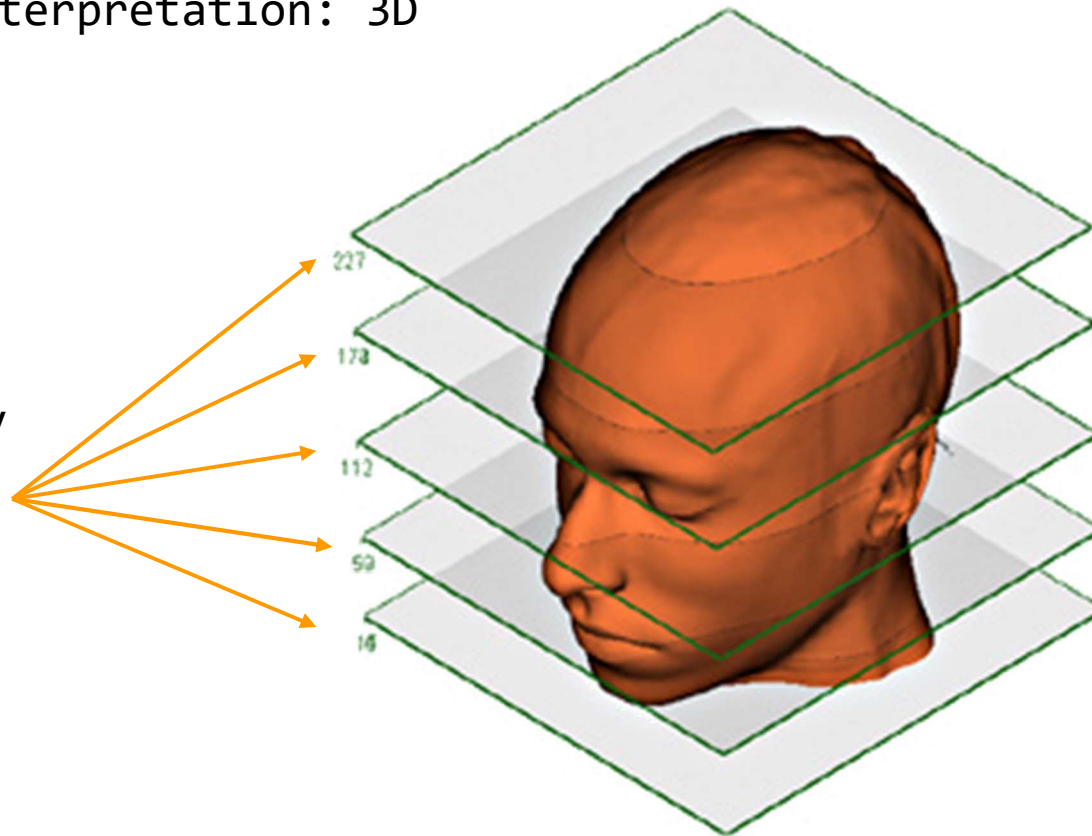
Input: perception

□ Visual

- Image
 - Interpretation: 3D

CT scanner slices up an object
into several layers

Computed
Tomography
Imaging
(CT Scan,
CAT Scan)



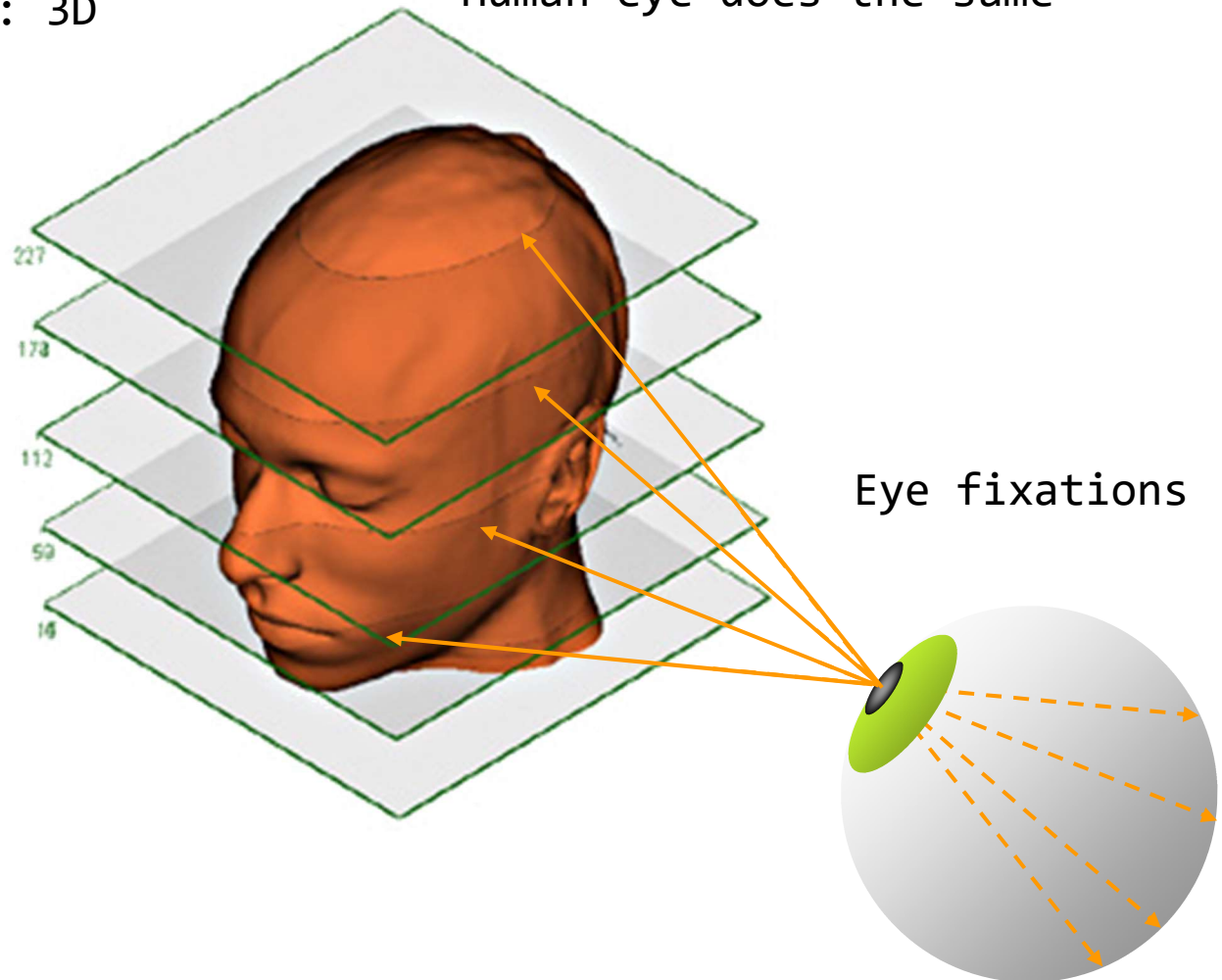
Input: perception

□ Visual

- Image
 - Interpretation: 3D

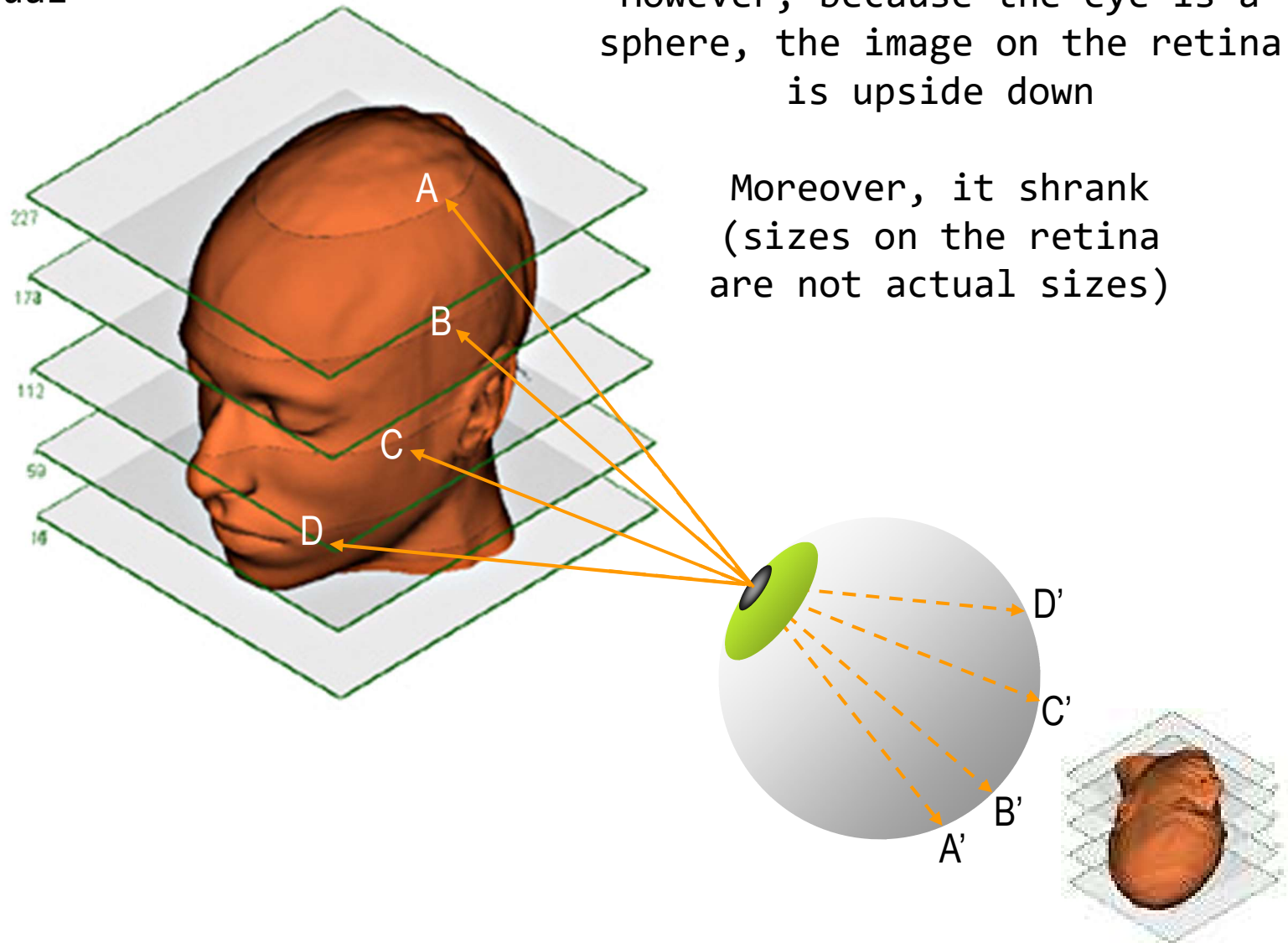
CT scans are slices of an object,
showing several layers
Human eye does the same

Missing data
(in between the
fixations) are
filled in
conceptually
(from memory)



Input: perception

❑ Visual



Input: perception

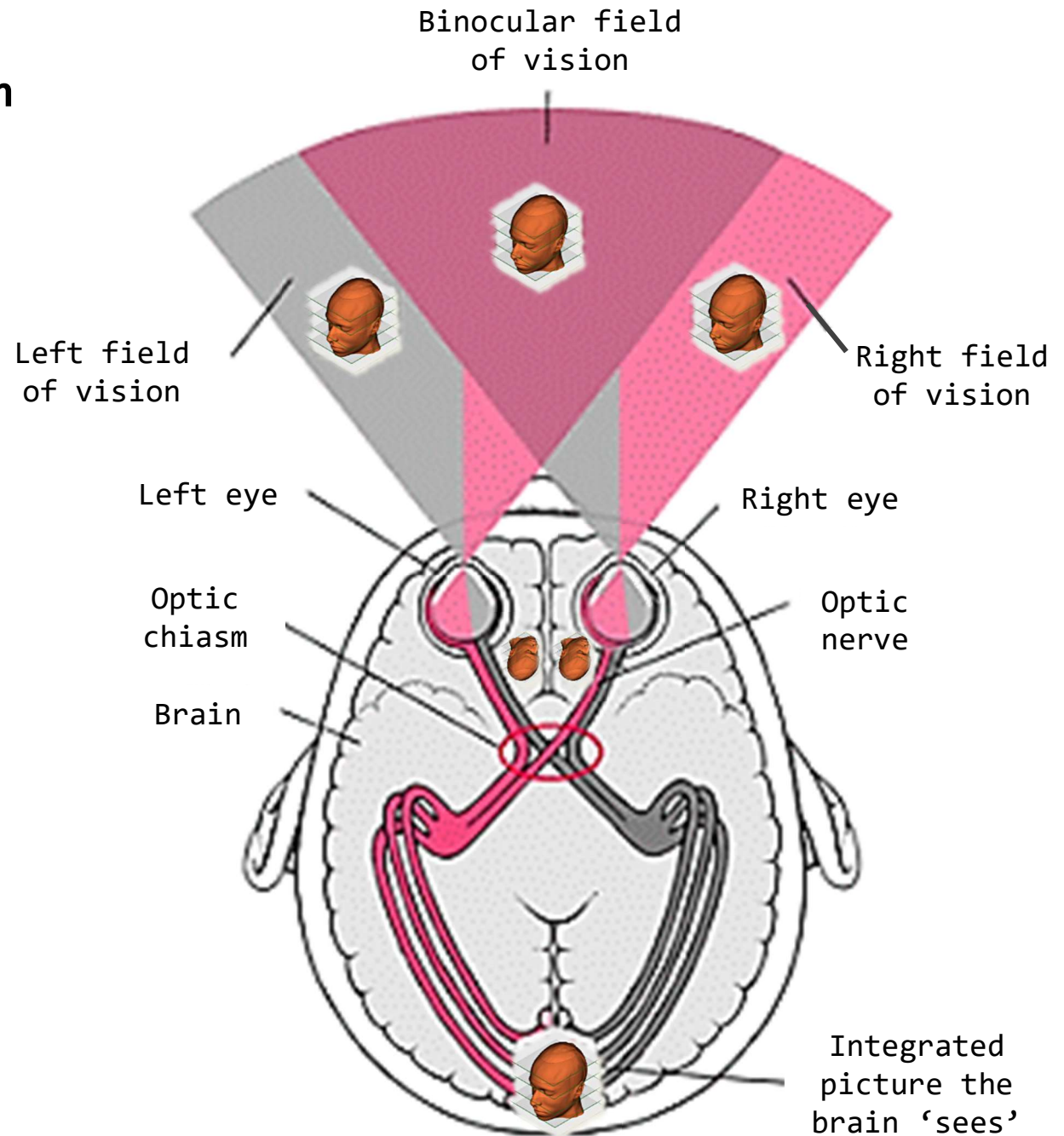
□ Visual

Mind you, you have two eyes
So there are two upside down images

These two images were taken from different angles

To merge them into one, they enter the optic chiasm

Images are split:
left goes right
right goes left



Input: perception

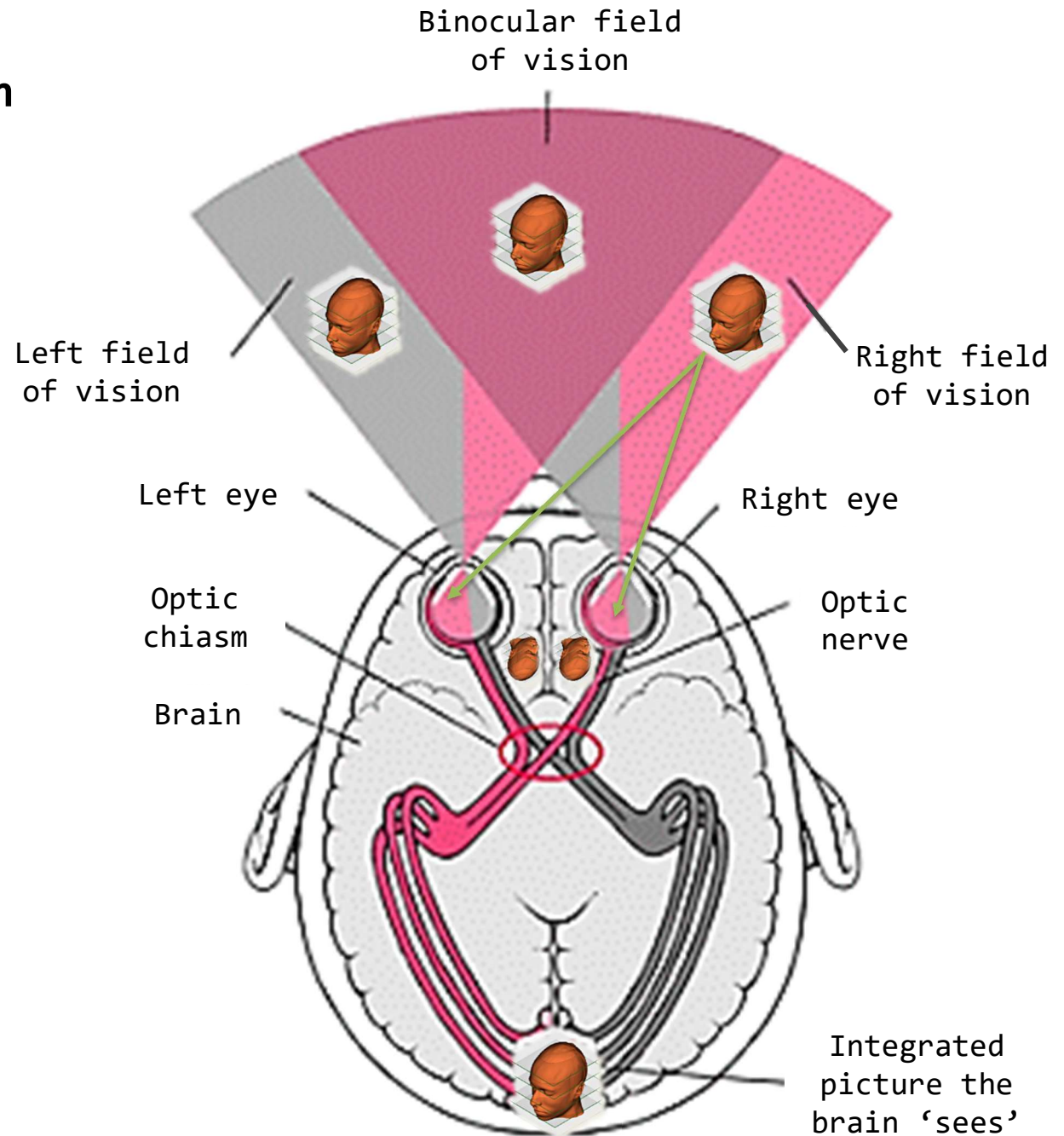
□ Visual

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Input: perception

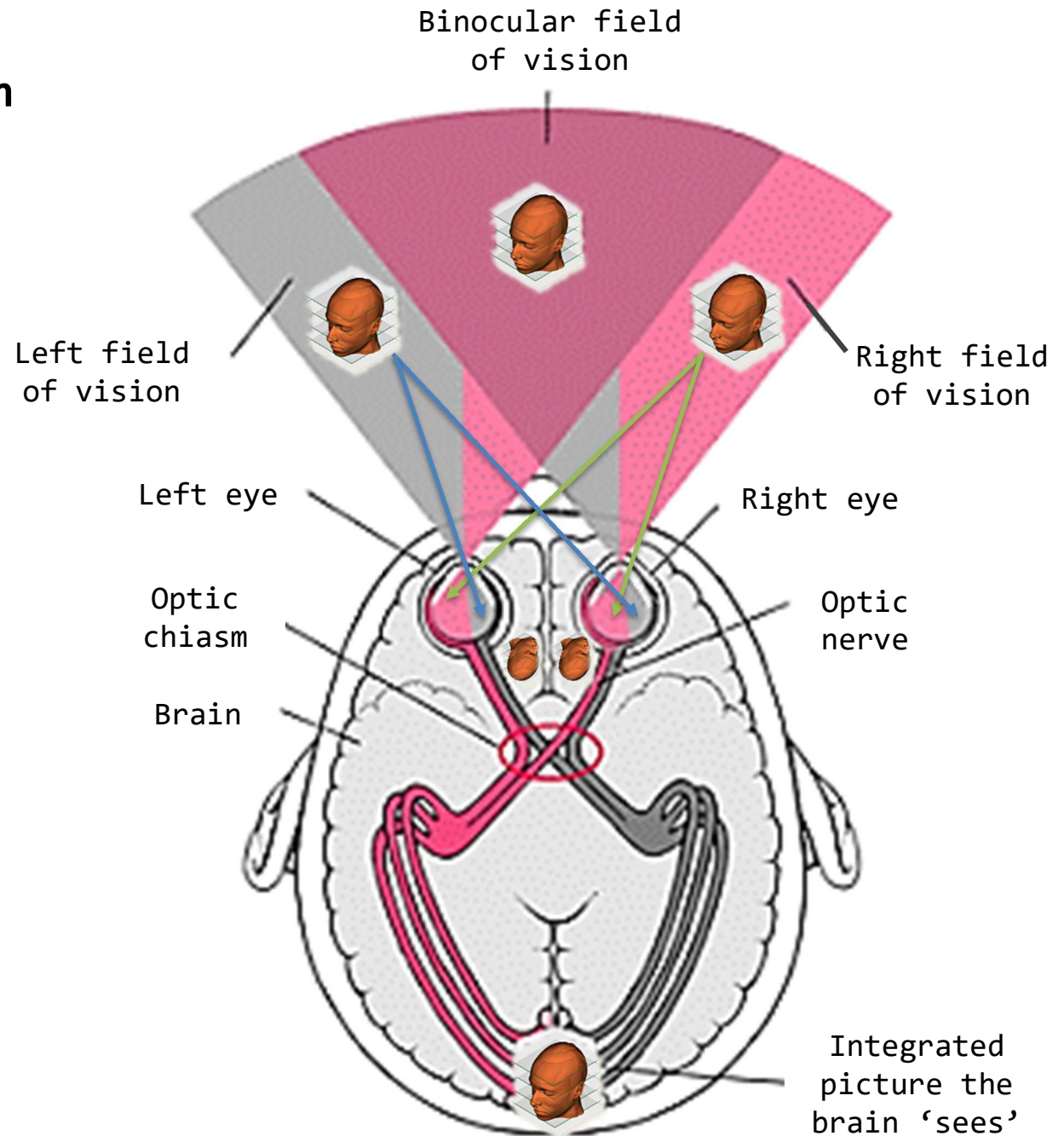
□ Visual

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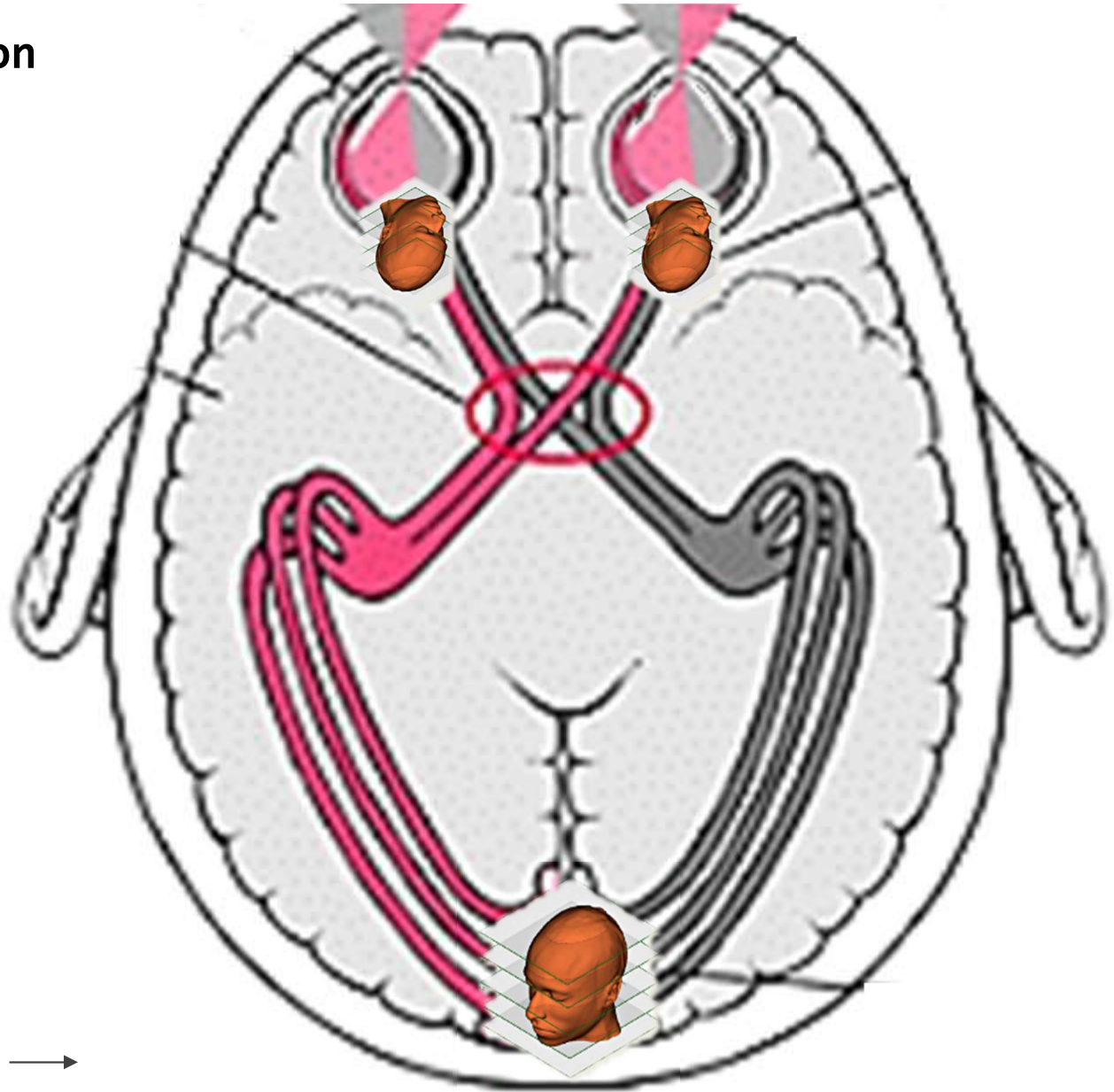
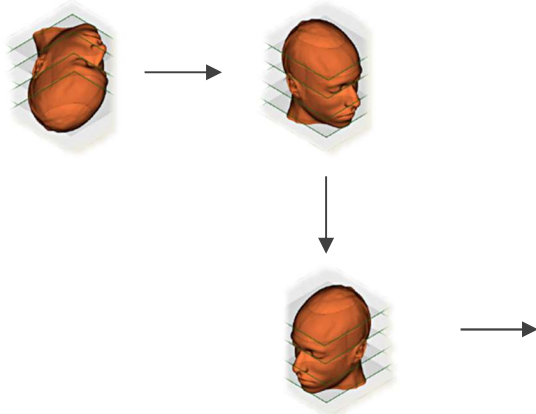


Input: perception

□ Visual

But before the image can be passed on from the visual cortex to the next station, it must be:

Flipped, rotated, and resized

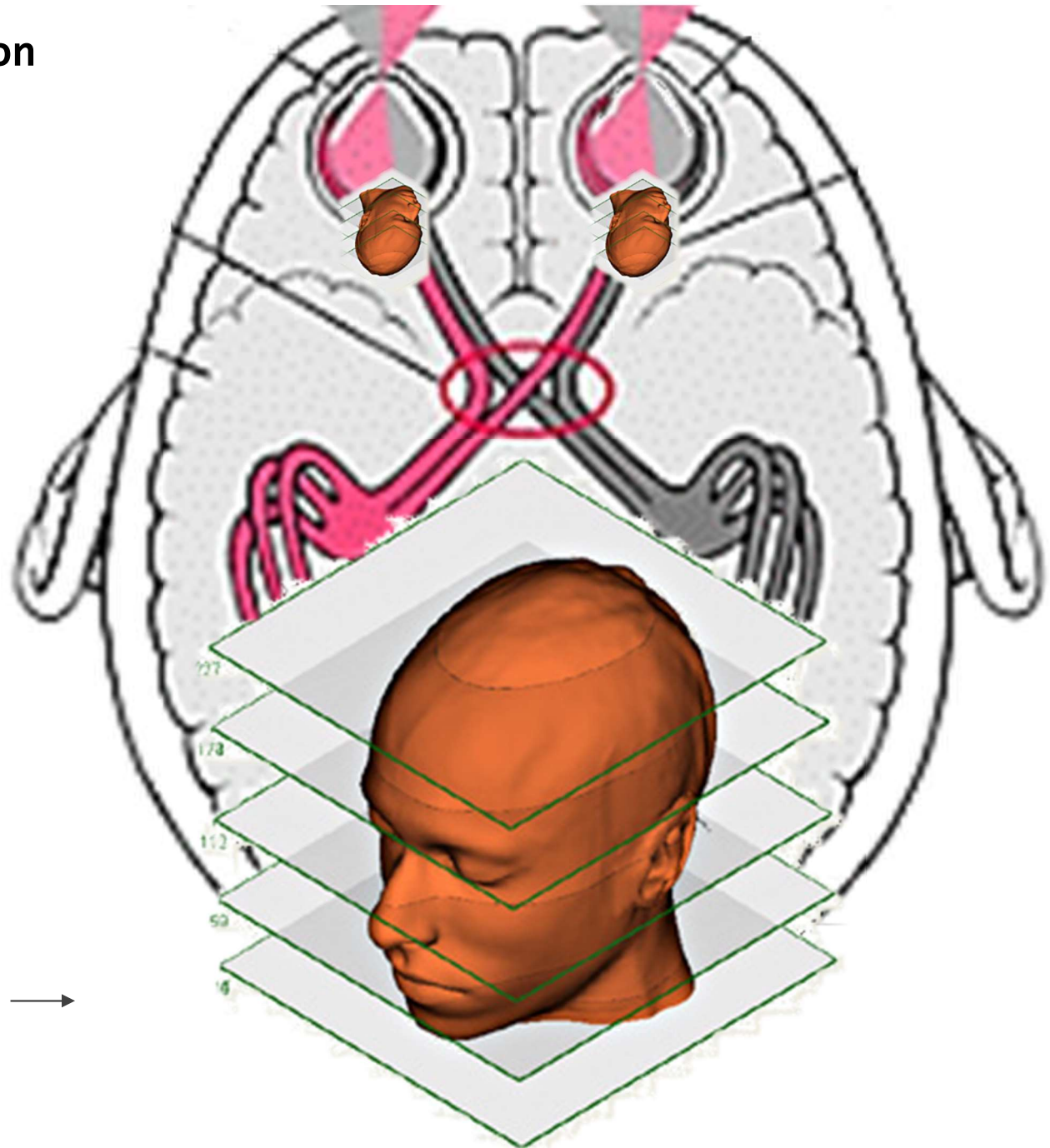
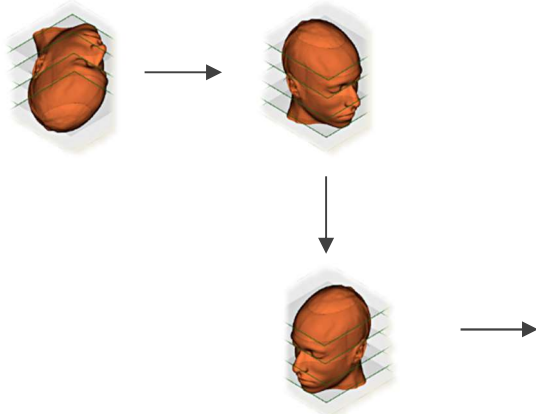


Input: perception

□ Visual

But before the image can be passed on from the visual cortex to the next station, it must be:

Flipped, rotated, and resized



Input: perception

❑ Visual

- Image
 - Interpretation: 3D



Therefore, 3D vision is **NO** property of image on the retina (which is 2D)

- Slicing up (eye fixations)
- Imputing missing data from memory (concept-driven percept)
- Shrinking images
- Taking them from different angles
- Splitting them up
- Left right, right left
- Merging
- Flip
- Rotate
- Resize (blow up)



Delvacam DELV-3DMS Dual 8 Inch LCD Monitor with Attached 3D Stereoscopic Viewer

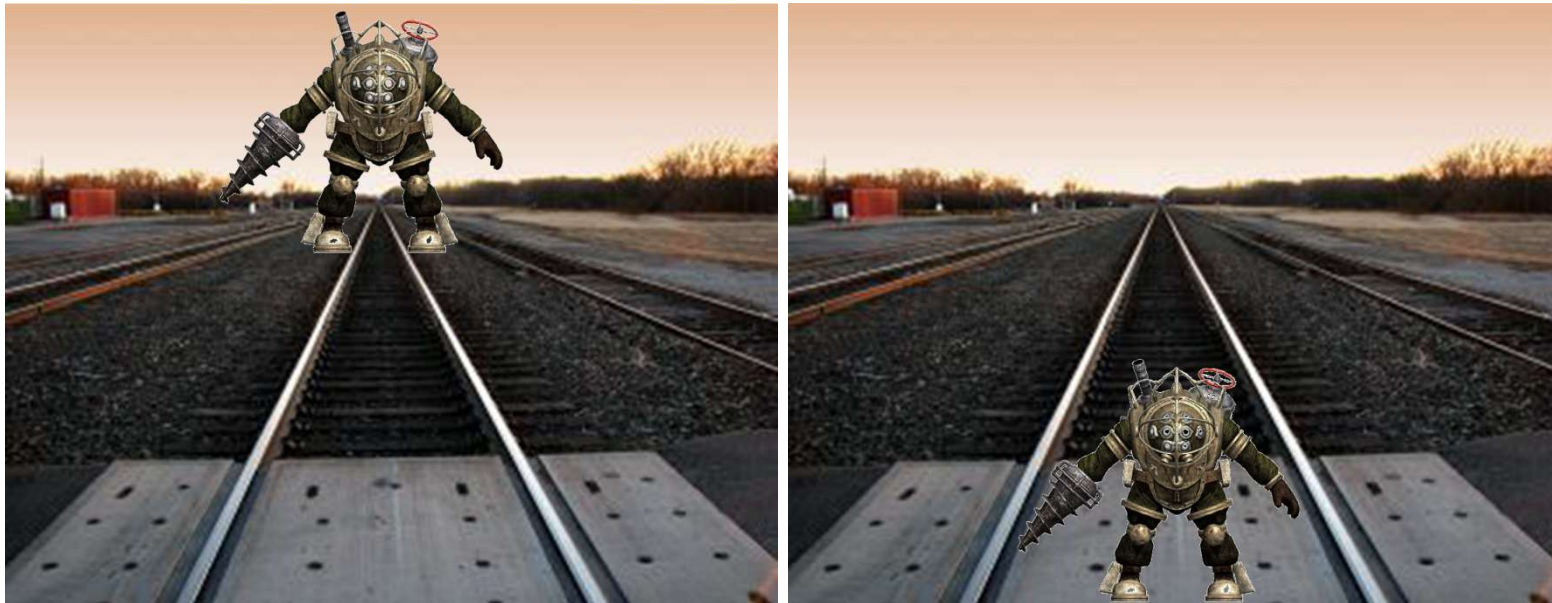
Input: perception

□ Visual

- Image
 - Interpretation: 3D

The illusion of depth is so strong that it urges our brain to resize objects so as to fit our conceptual interpretation of the environment

Size and depth



Giant or gnome?

Slanted lines indicate size through depth suggestion

Input: perception

□ Visual

- Image
 - Interpretation: 3D

Size and depth

- Visual angle indicates how much of view is occupied by the object
- Related to size and distance from eye
- Visual acuity is ability to perceive fine detail
- 'Law of size constancy' means that familiar objects are perceived as constant size in spite of changes in visual angle
- Cues like 'overlapping' help perception of size and depth

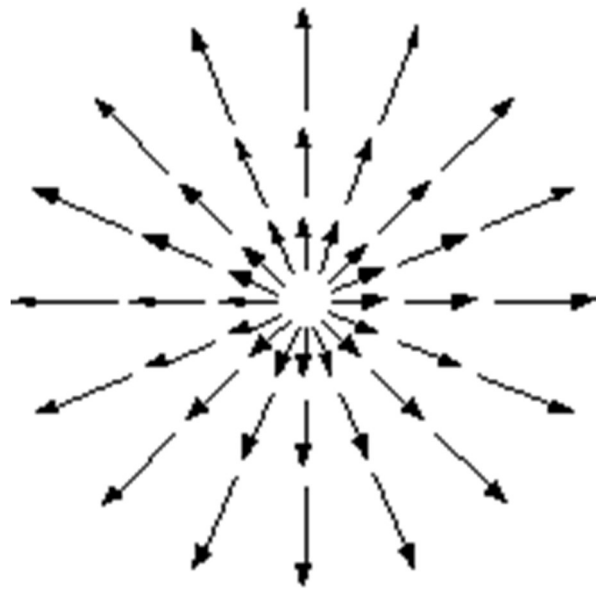


Input: perception

□ Visual

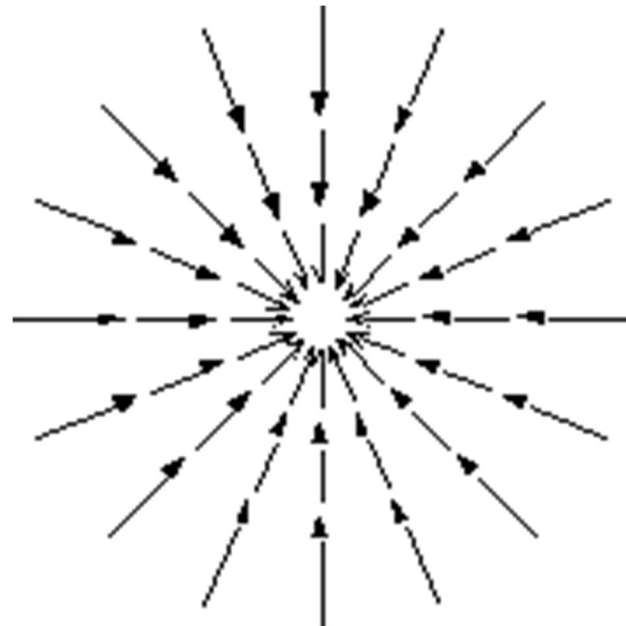
- Image
 - Interpretation: 3D

Positioning through motion



Expanding optic flow makes the body lean forward

Optic flow is the brain's cue to calculate position in space



Contracting optic flow makes the body lean backward

Input: perception

□ Visual

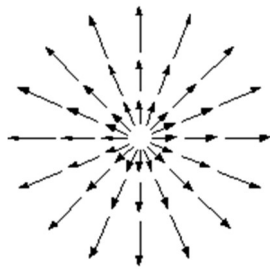
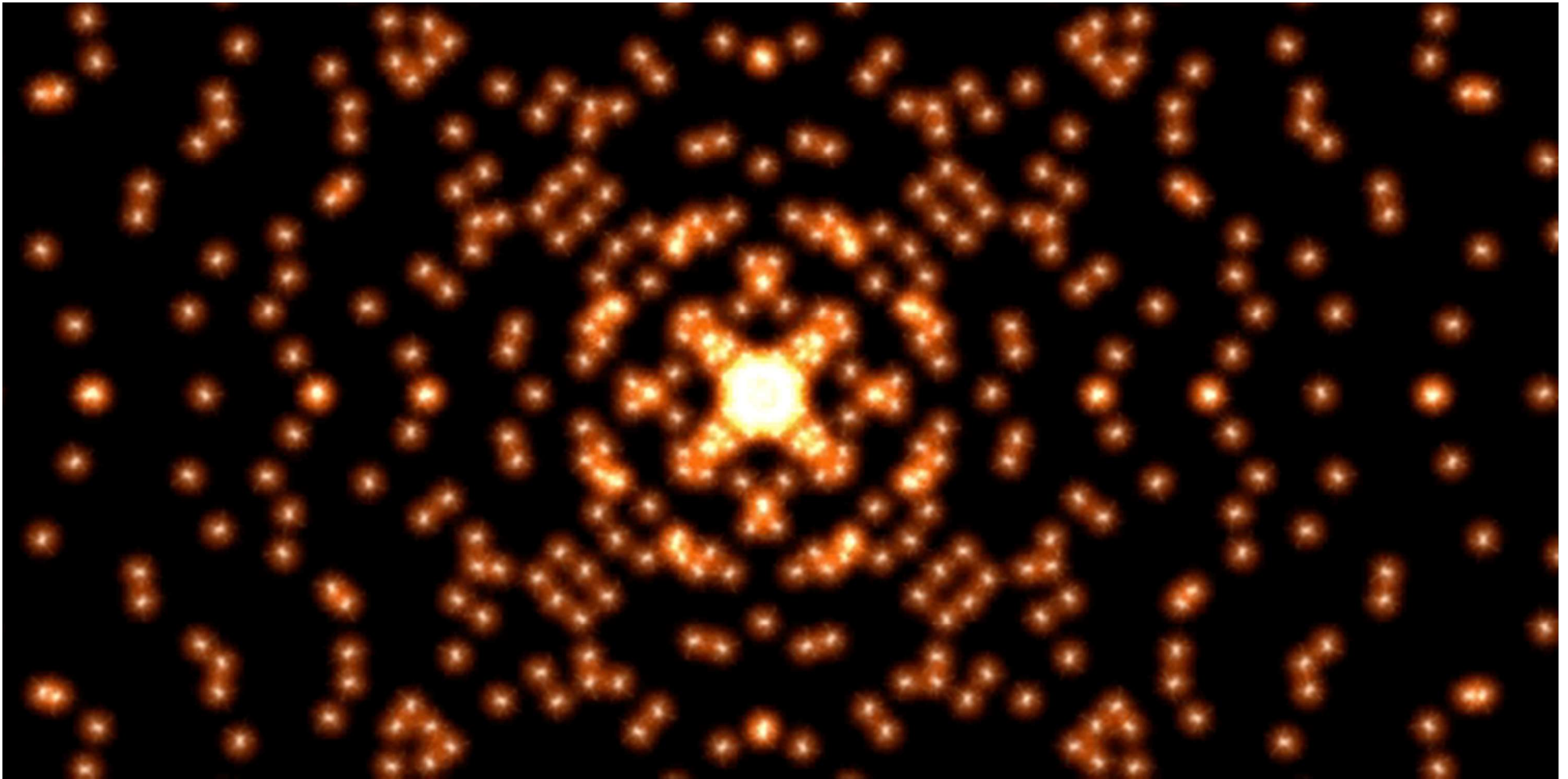
- Image
 - Interpretation: 3D

Optic flow is the brain's cue to calculate position in space

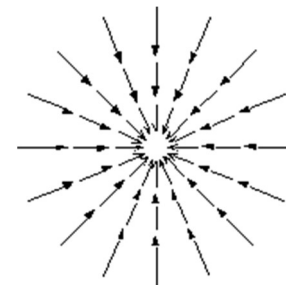
Positioning through motion



Expanding optic flow makes the body lean forward



Expanding optic flow makes the
body lean forward



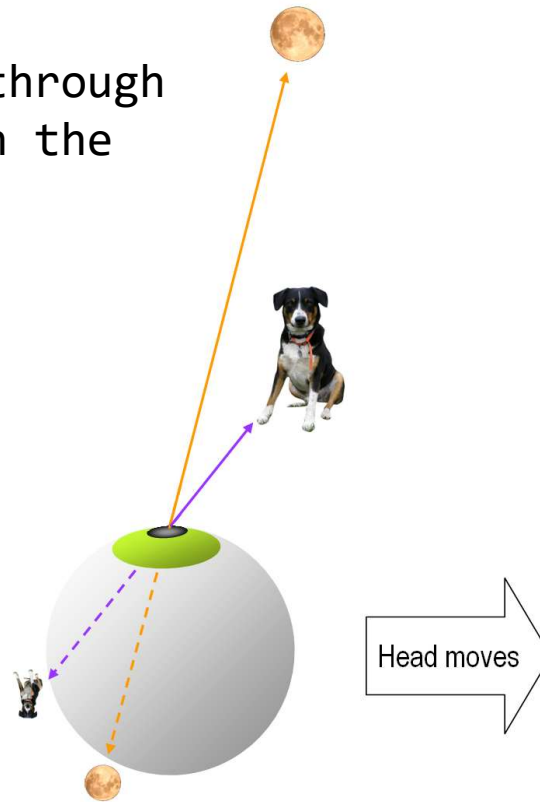
Contracting optic flow makes
the body lean backward

Input: perception

□ Visual

- Image
 - Interpretation: 3D

Seeing 3D through motion when the head moves



Dog 'moves' more on the retina than the moon because the length of the projection path of the dog on the retina is longer (purple arrow) than that of the moon (orange arrow)

The brain interprets this as: Short path is behind long path; therefore, there must be depth

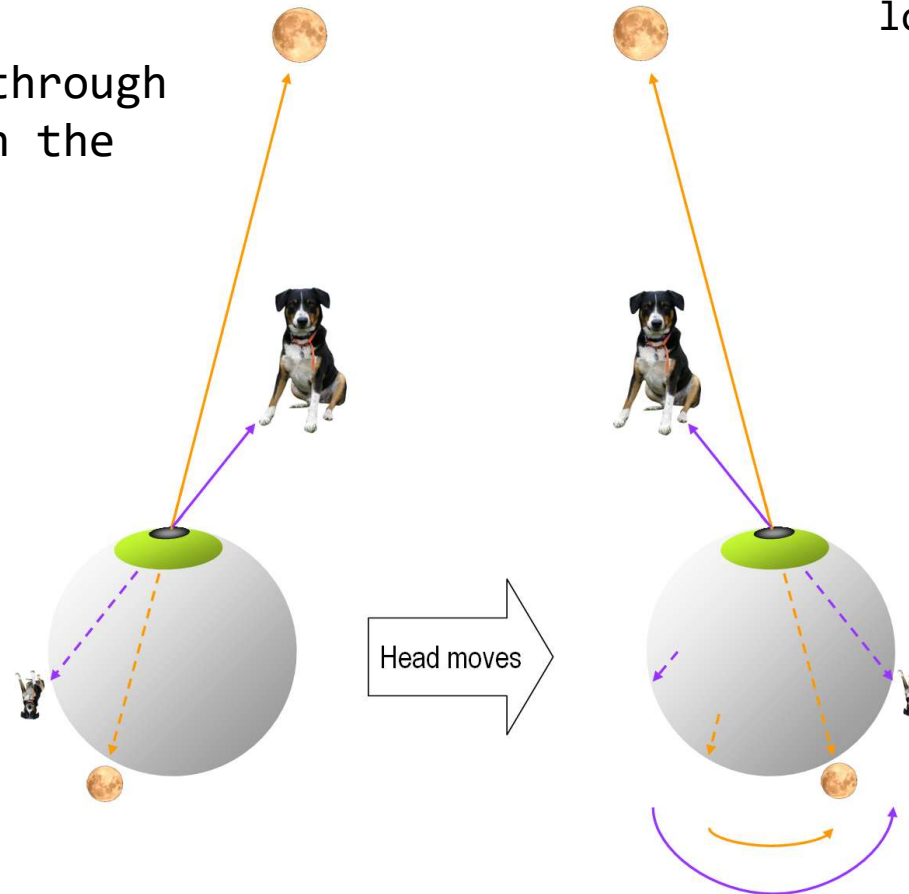
Input: perception

□ Visual

- Image

- Interpretation: 3D

Seeing 3D through motion when the head moves



Dog 'moves' more on the retina than the moon because the length of the projection path of the dog on the retina is longer (purple arrow) than that of the moon (orange arrow)

The brain interprets this as: Short path is behind long path; therefore, there must be depth

Input: perception

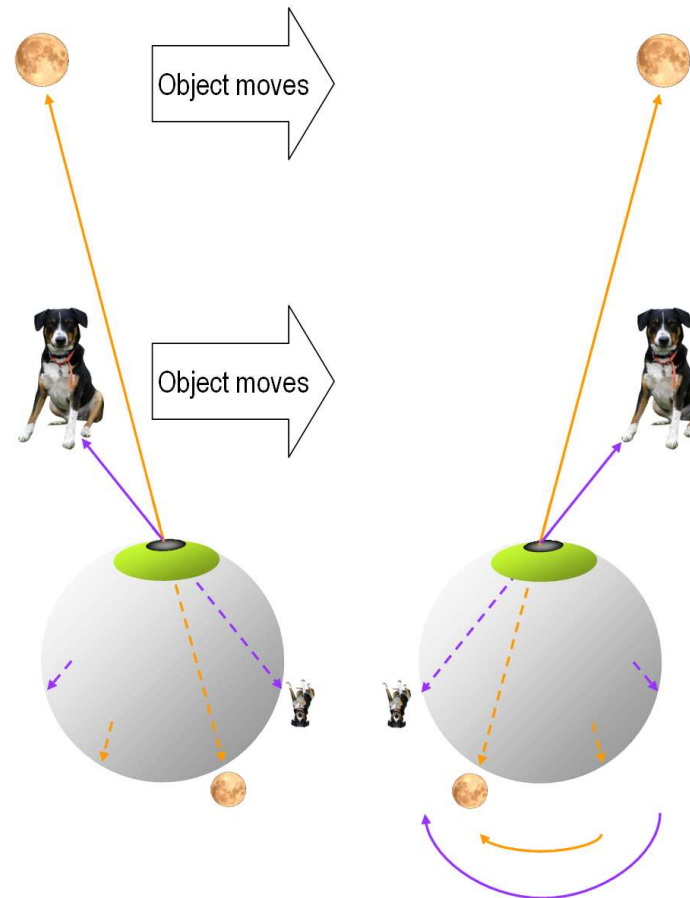
□ Visual

- Image
 - Interpretation: 3D

Seeing 3D through motion when the object moves

Therefore,
'Moving-train illusion'
It's the other train that's moving

The object closest to the eye 'moves' the most because the retinal path is longer. The brain interprets this as: Short path is behind long path; therefore, there must be depth

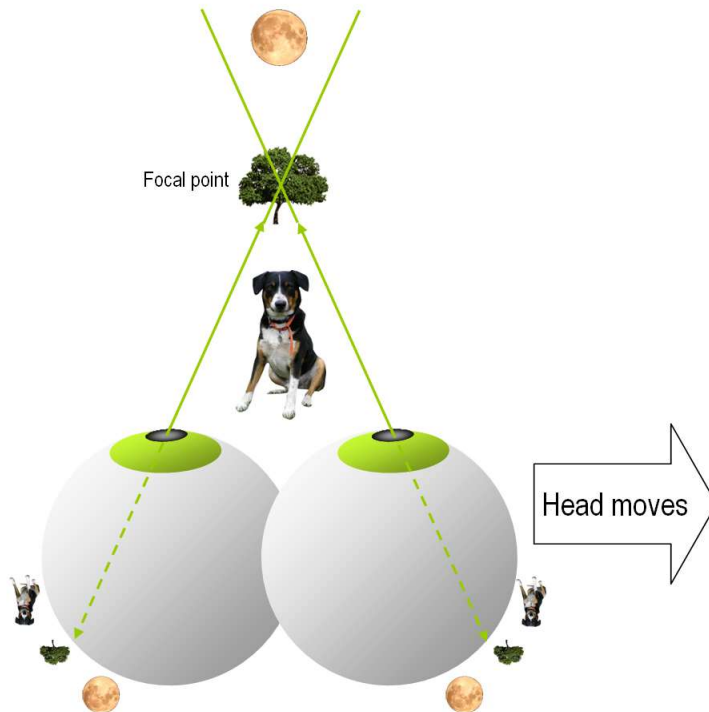


Input: perception

□ Visual

- Image
 - Interpretation: 3D

Motion parallax



The direction in which an object leaves the visual field indicates the direction of object motion in the physical world. Moon travels with you (which may not be true)

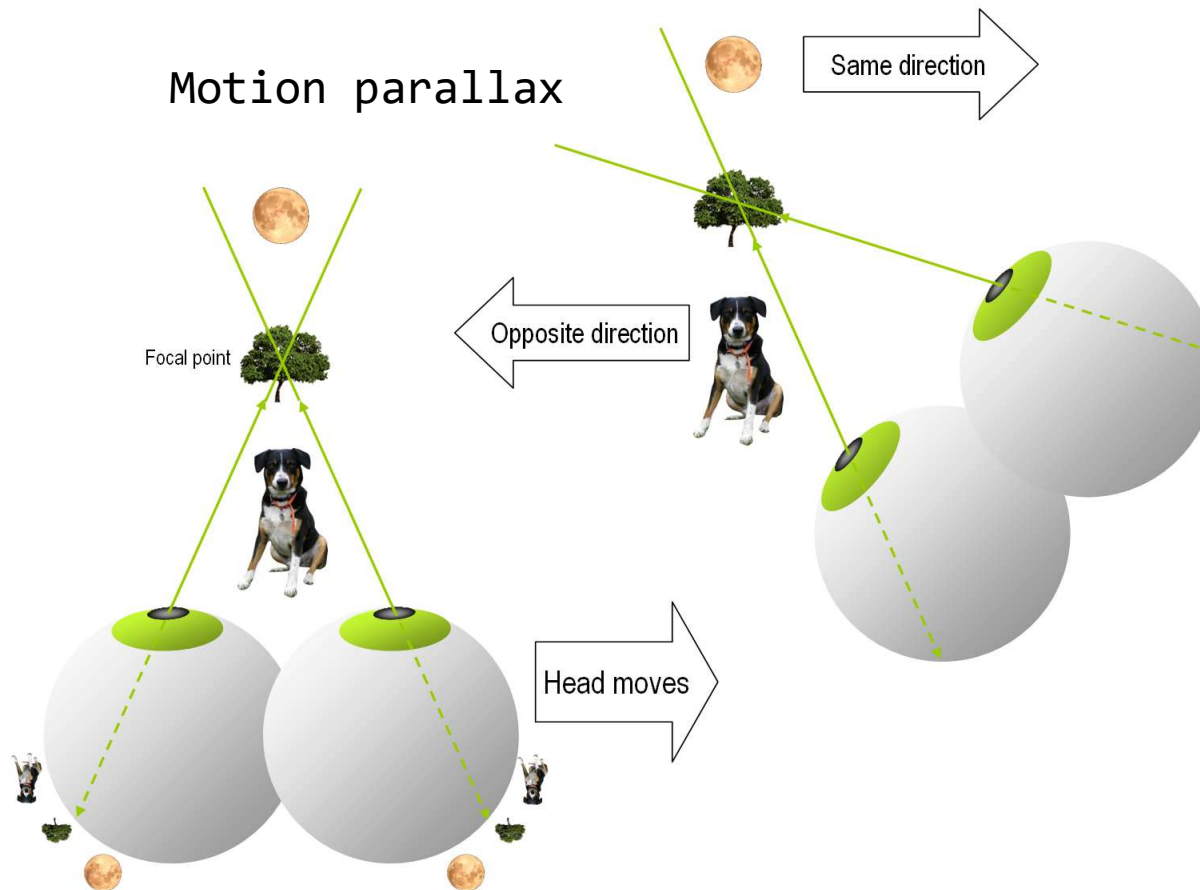
Thus, direction of leaving the visual half-field cues object motion and despite conceptually knowing that the moon does not fly with you, the simple everyday heuristics make us see it that way

Input: perception

□ Visual

- Image
 - Interpretation: 3D

Motion parallax



The direction in which an object leaves the visual field indicates the direction of object motion in the physical world. Moon travels with you (which may not be true)

Thus, direction of leaving the visual half-field cues object motion and despite conceptually knowing that the moon does not fly with you, the simple everyday heuristics make us see it that way

Input: perception

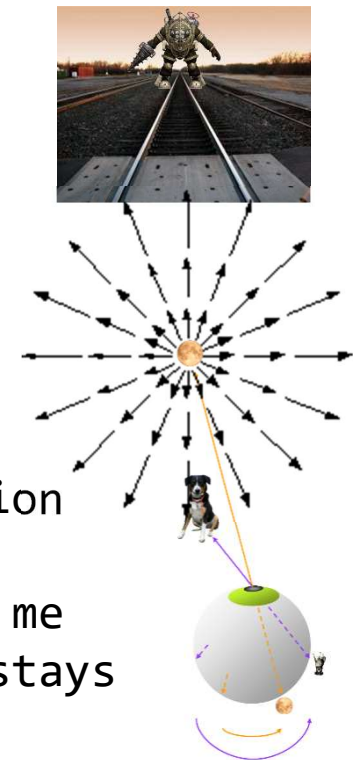
□ Visual

- Image
 - Interpretation: 3D



Therefore, 3D vision is **NO** property of image on the retina (which is 2D)

- Visual angle indicates object size
- Related to size and distance from eye
- Size constancy despite changes in visual angle
- Expanding optic flow indicates 'we go forward'
- Contracting optic flow indicates 'backward'
- Long retinal path indicates 'close by'
- Short retinal path indicates 'far away'
- Side of leaving visual half field indicates motion direction:
 - side of your direction means object goes with me
 - opposite side of your direction means object stays behind



So what is 3D about this?



Or this?



Overwatch

Or this?



Next Car Game

Short break, after that:
Pattern recognition

Pattern recognition

Input: perception

☐ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Chin Ho	8	1	5	1	9	1	7	1	
3	Po Ki	2	3	1	2	2	3	1	2	
4	King to	7	1	9	1	7	1	8	2	
5	Jiayun	1	2	2	3	1	3	2	3	
6	Yu Heng	5	1	7	1	9	2	9	1	
7	Jin	1	2	1	3	2	3	2	3	
8	Genggeng	9	2	9	2	8	1	6	1	
9	Mingyi	2	3	1	2	1	2	2	3	
10	Yuyang	8	1	9	1	8	2	4	1	
11	Yajie	1	2	1	3	1	3	2	3	
12	Yifan	8	2	7	1	7	1	9	1	
13	Xianghui	1	3	2	3	1	2	1	3	
14										
15										

Which is higher, which is lower?

Input: perception

☐ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Chin Ho	8	1	5	1	9	1	7	1	
3	Po Ki	2	3	1	2	2	3	1	2	
4	King to	7	1	9	1	7	1	8	2	
5	Jiayun	1	2	2	3	1	3	2	3	
6	Yu Heng	5	1	7	1	9	2	9	1	
7	Jin	1	2	1	3	2	3	2	3	
8	Genggeng	9	2	9	2	8	1	6	1	
9	Mingyi	2	3	1	2	1	2	2	3	
10	Yuyang	8	1	9	1	8	2	4	1	
11	Yajie	1	2	1	3	1	3	2	3	
12	Yifan	8	2	7	1	7	1	9	1	
13	Xianghui	1	3	2	3	1	2	1	3	
14		high	low	high	low	high	low	high	low	
15										

Which is higher, which is lower?

Input: perception

□ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Chin Ho	high	low	high	low	high	low	high	low	
3	Po Ki	low	high	low	high	low	high	low	high	
4	King to	high	low	high	low	high	low	high	low	
5	Jiayun	low	high	low	high	low	high	low	high	
6	Yu Heng	high	low	high	low	high	low	high	low	
7	Jin	low	high	low	high	low	high	low	high	
8	Genggeng	high	low	high	low	high	low	high	low	
9	Mingyi	low	high	low	high	low	high	low	high	
10	Yuyang	high	low	high	low	high	low	high	low	
11	Yajie	low	high	low	high	low	high	low	high	
12	Yifan	high	low	high	low	high	low	high	low	
13	Xianghui	low	high	low	high	low	high	low	high	
14										

Which is higher, which is lower?

Input: perception

□ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G	H	I	J
1		Apple1	MS1	Apple2	MS2	Apple3	MS3	Apple4	MS4	
2	Male	high	low	high	low	high	low	high	low	
3	Female	low	high	low	high	low	high	low	high	
4	Male	high	low	high	low	high	low	high	low	
5	Female	low	high	low	high	low	high	low	high	
6	Male	high	low	high	low	high	low	high	low	
7	Female	low	high	low	high	low	high	low	high	
8	Male	high	low	high	low	high	low	high	low	
9	Female	low	high	low	high	low	high	low	high	
10	Male	high	low	high	low	high	low	high	low	
11	Female	low	high	low	high	low	high	low	high	
12	Male	high	low	high	low	high	low	high	low	
13	Female	low	high	low	high	low	high	low	high	
14										

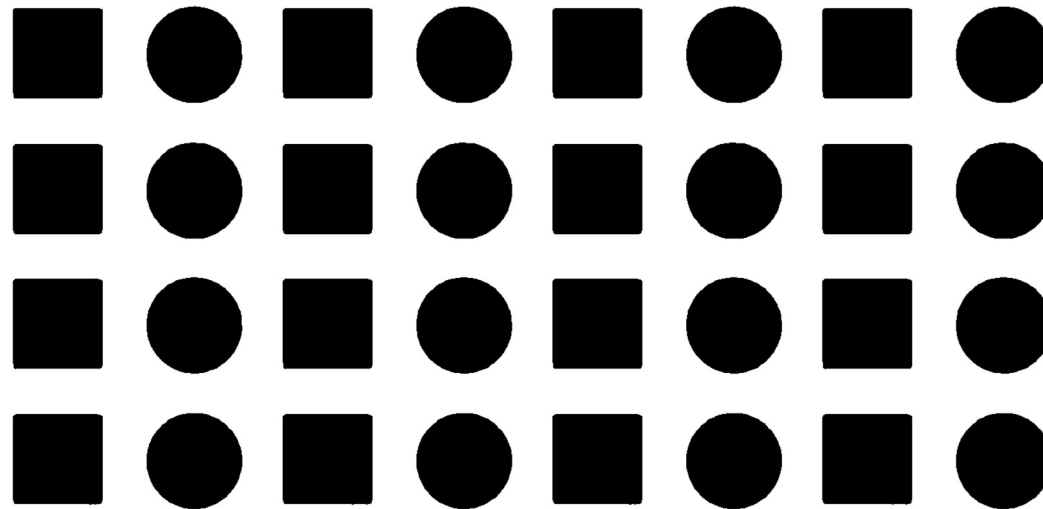
Which is higher, which is lower?

Input: perception

□ Visual

Items that are similar tend
to be grouped together

- Image
 - Pattern recognition: similarity



Most people will see columns of
squares and circles, rather than
rows of alternating shapes

Input: perception

□ Visual

- Image
 - Pattern recognition

	A	B	C	D	E	F	G
1		var1	var2	var3	var4	var5	
2	record1	3	3	7	7	7	
3	record2	3	3	7	7	7	
4	record3	3	3	7	7	7	
5	record4	3	3	0	0	0	
6	record5	3	3	0	0	0	
7	record6	3	3	0	0	0	
8	record7	3	3	0	0	0	
9							
10							

Do you see any grouping in
this data set?

Input: perception

□ Visual

- Image
 - Pattern recognition

Humans tend to
recognize patterns even
where they are not

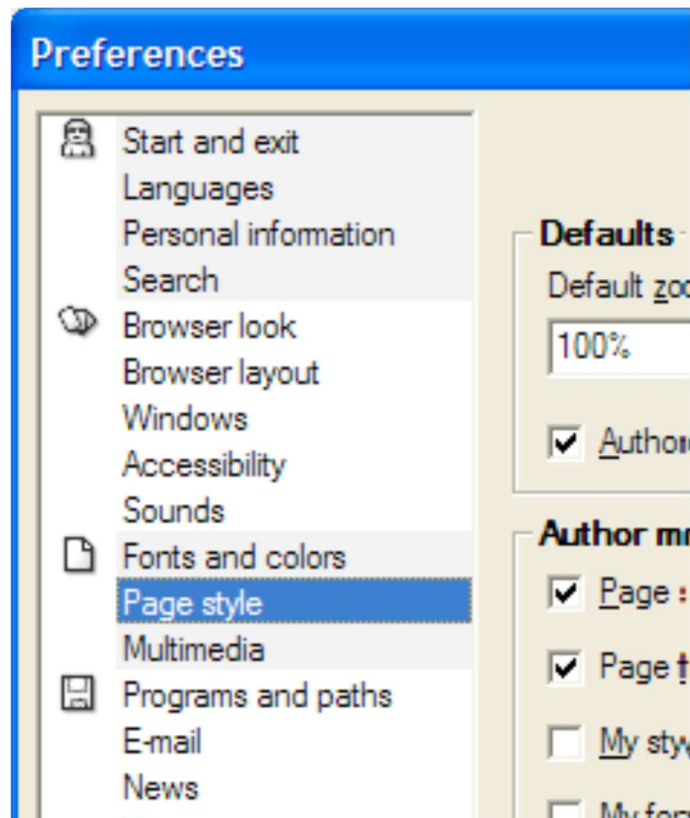
	A	B	C	D	E	F	G
1		var1	var2	var3	var4	var5	
2	record1	3	3	7	7	7	
3	record2	3	3	7	7	7	
4	record3	3	3	7	7	7	
5	record4	3	3	0	0	0	
6	record5	3	3	0	0	0	
7	record6	3	3	0	0	0	
8	record7	3	3	0	0	0	
9							
10							

Probably you see three groups of
data points although these groups
are unrelated

Input: perception

□ Visual

- Image
 - Pattern recognition: similarity



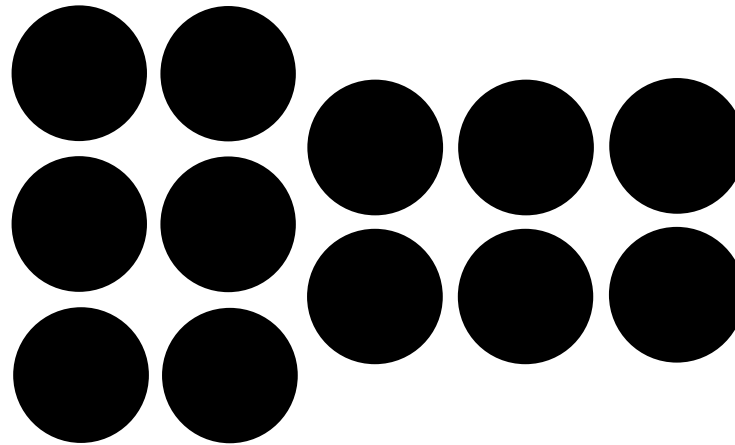
From the Opera browser.
Options with the same
background color are
“tied together”
automatically by the user

Input: perception

□ Visual

Objects near each other tend
to be grouped together

- Image
 - Pattern recognition: proximity



We will most likely see the circles on
the left as grouped into two columns,
and those on the right as grouped into
two rows

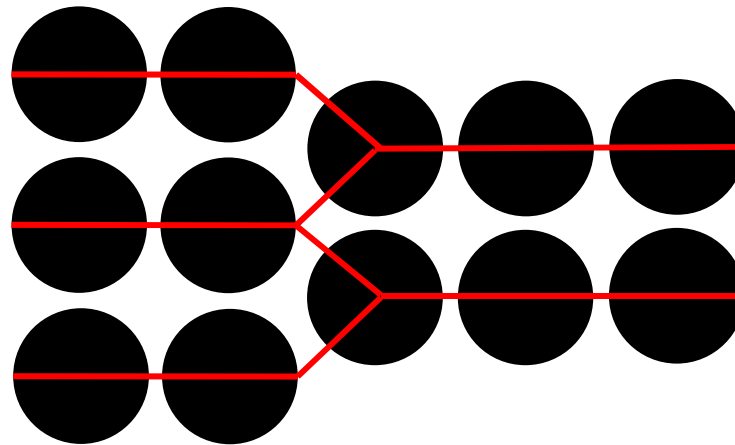
Input: perception

□ Visual

Objects near each other tend to be grouped together

- Image

- Pattern recognition: proximity



You probably did not see this one

We will most likely see the circles on the left as grouped into two columns, and those on the right as grouped into two rows

Input: perception

☐ Visual

- Image
 - Pattern recognition: proximity



☒ P2P Search ☐ Web Search

☒ Everything

☐ Audio

☐ Video

☐ Images

☐ Documents

☐ Software

☐ Playlists

Input: perception

□ Visual

- Image
 - Pattern recognition: symmetry



How many objects?

Input: perception

□ Visual

- Image

- Pattern recognition: symmetry

Symmetrical images are
perceived collectively, in
spite of distance

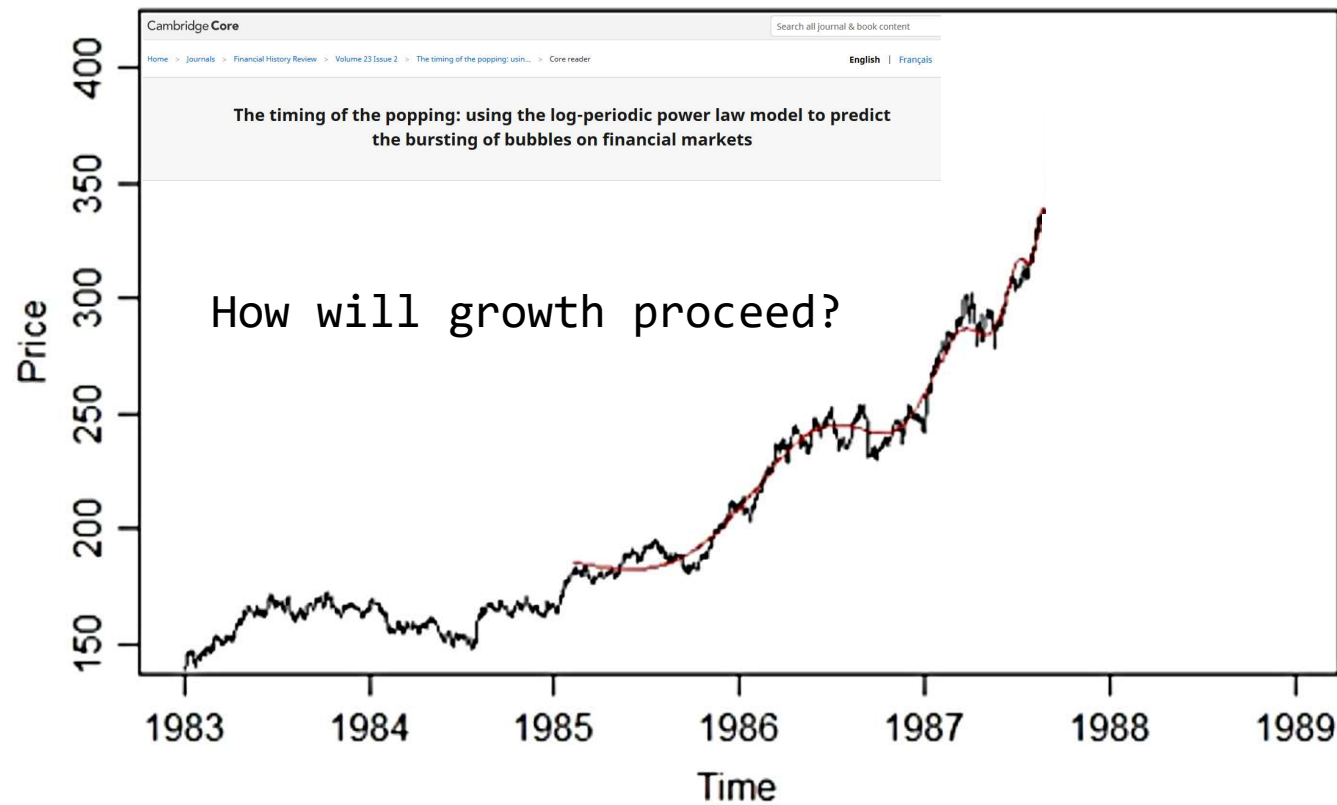


In the rectangle, we will most likely perceive three black
symmetrical shapes against a white background
(rather than four asymmetrical white objects against a
black background)

Input: perception

□ Visual

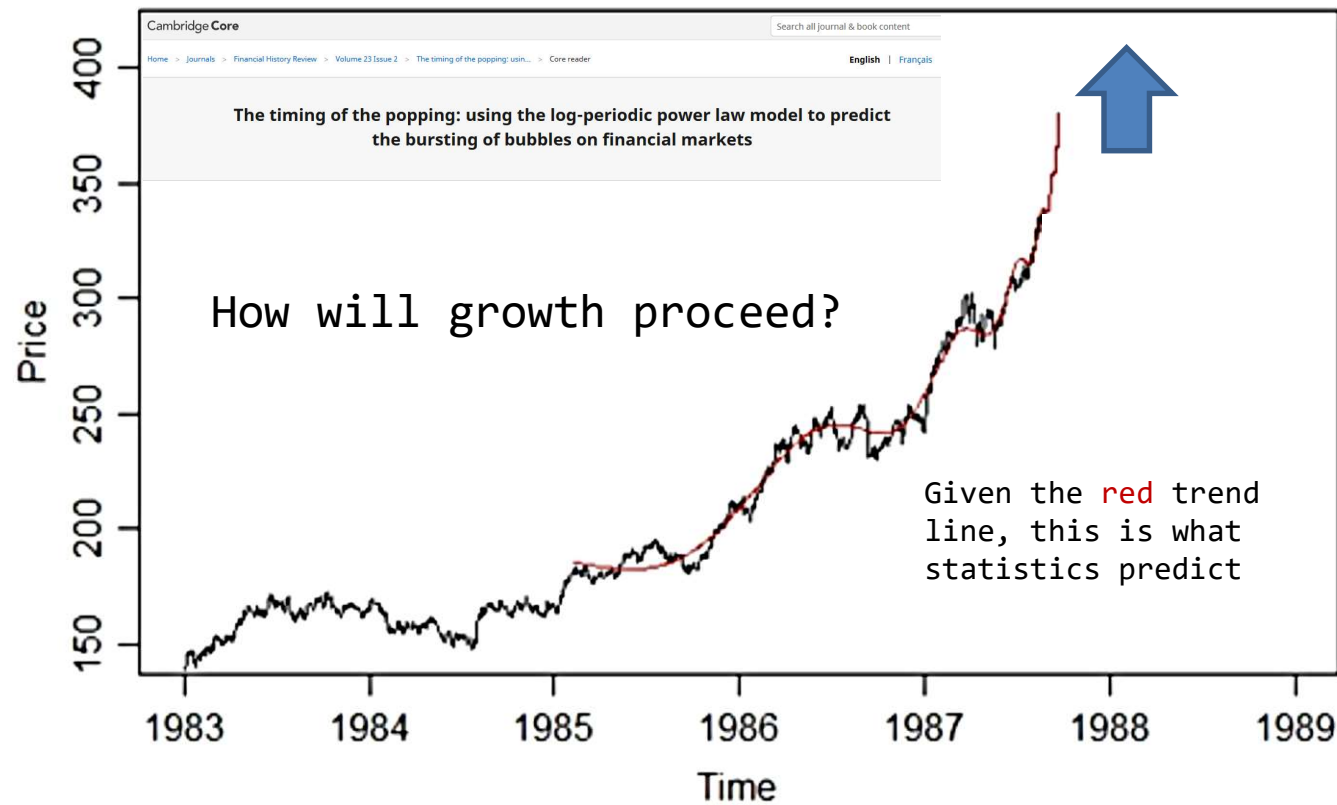
- Image
 - Pattern recognition: continuity



Input: perception

□ Visual

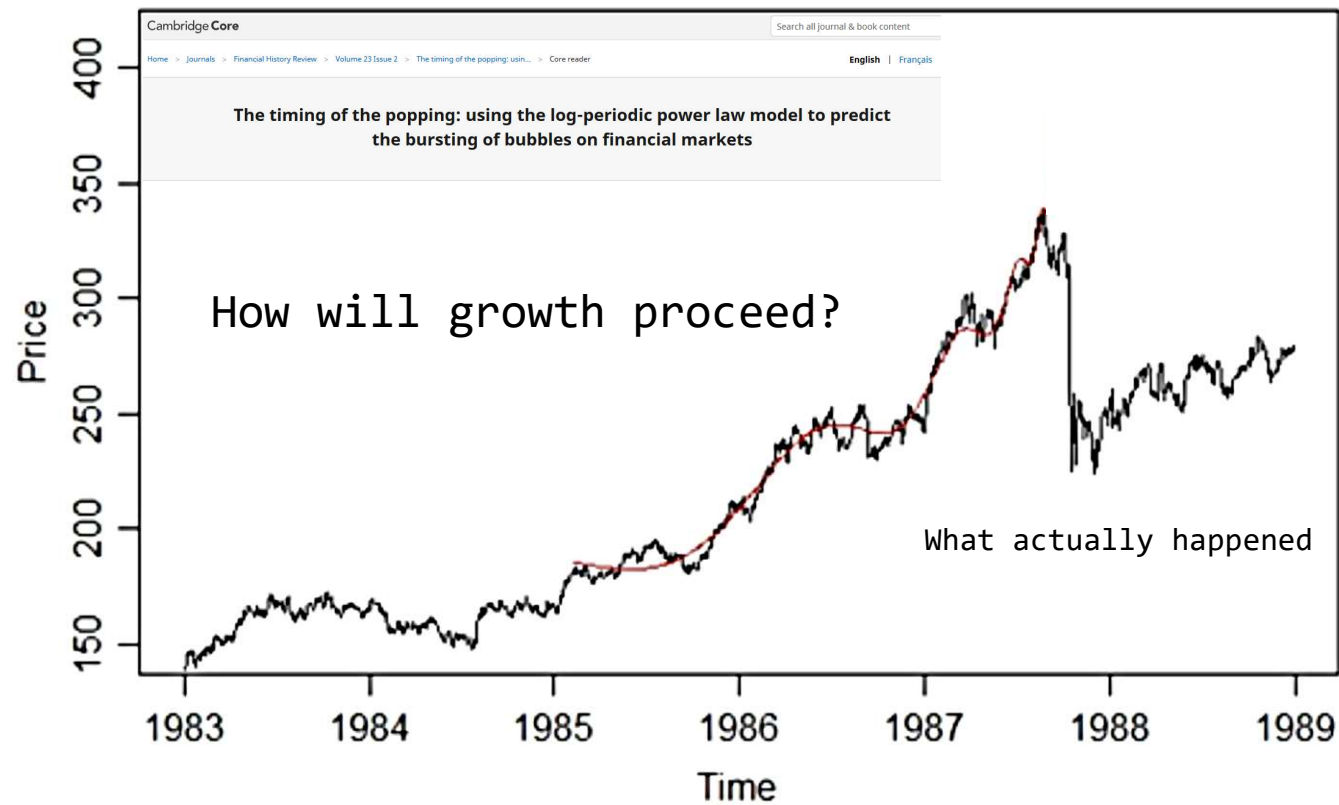
- Image
 - Pattern recognition: continuity



Input: perception

□ Visual

- Image
 - Pattern recognition: continuity

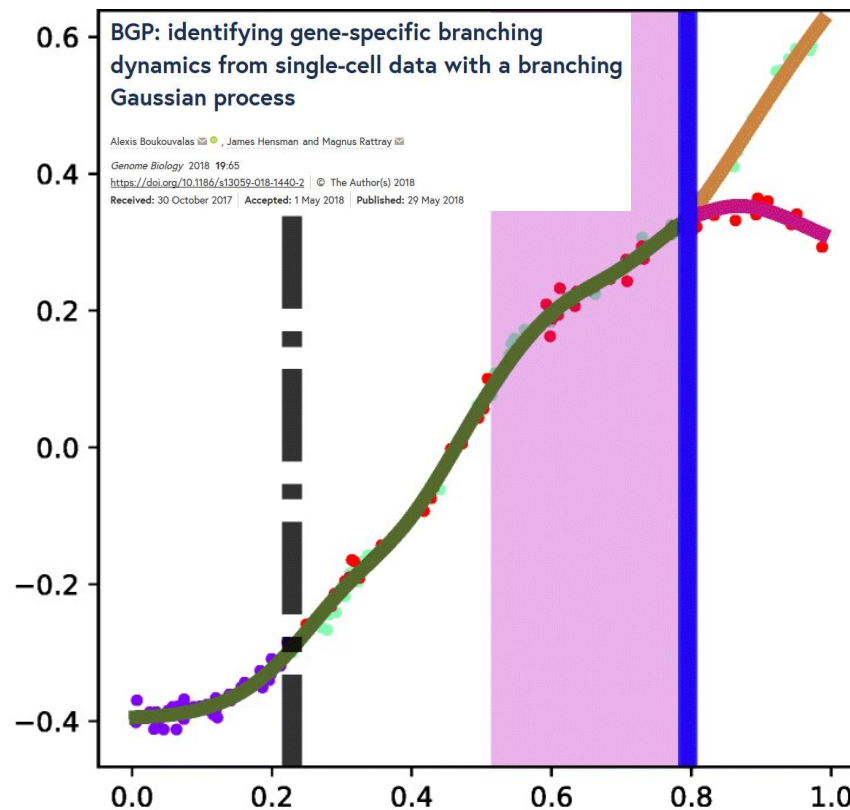


Input: perception

□ Visual

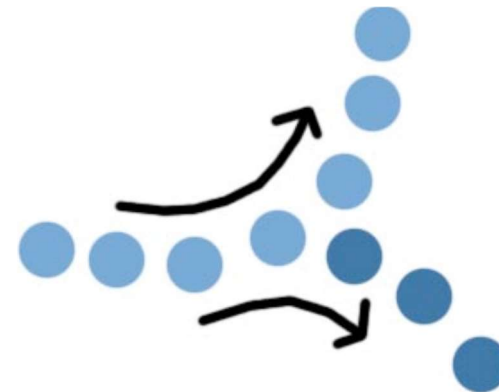
Lines are seen as following the smoothest path

- Image
 - Pattern recognition: continuity



(b) BGP Low noise

The top branch is seen as continuing the top segment of the line, rather than as an independent line



Input: perception

□ Visual

- Image
 - Pattern recognition: closure



How many objects?

Input: perception

□ Visual

- Image
 - Pattern recognition: closure

Objects grouped together are seen as a whole. We tend to ignore gaps and complete lines

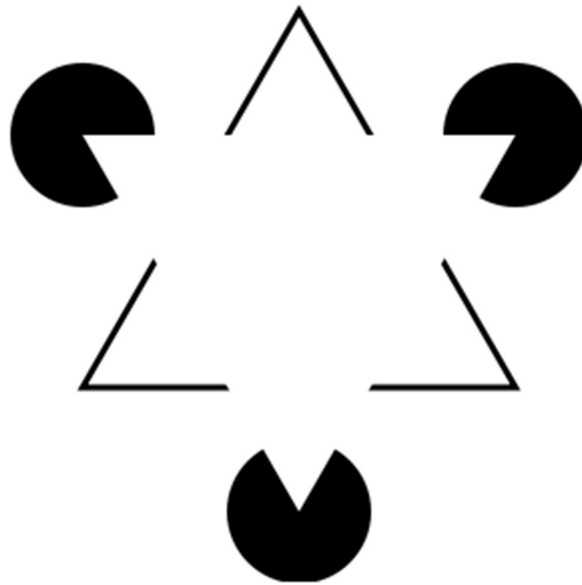


How many objects?



Blue lines on a white background.
Or? A blue background with slits
in the white paper?

How many triangles?
How many circles?
How many stars?

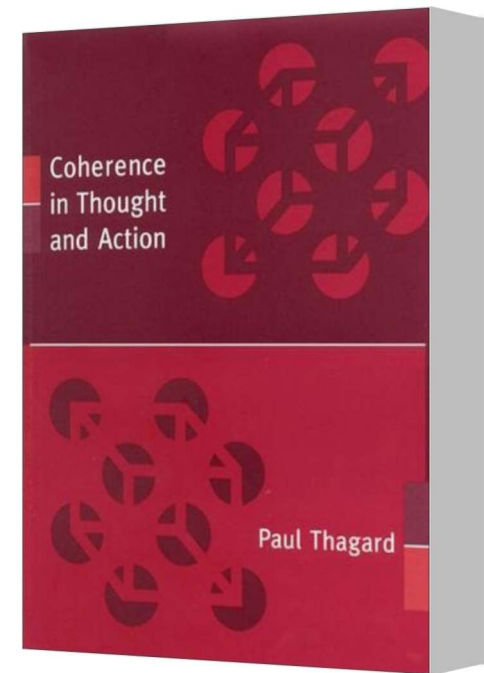
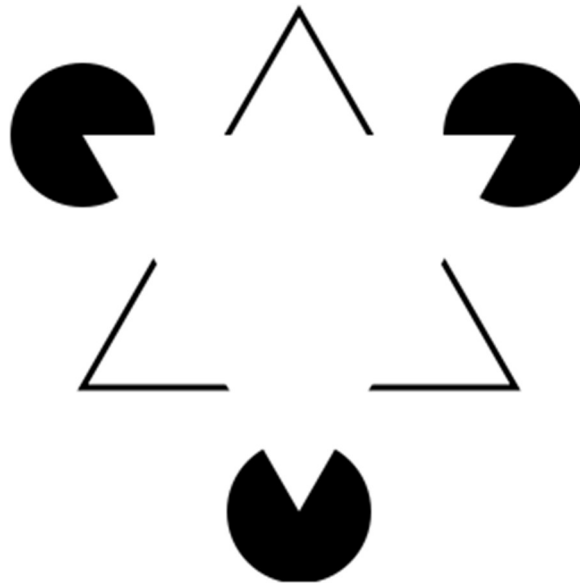


Kanizsa triangle

How many triangles?
How many circles?
How many stars?

There are no
circles or
triangles in
this picture.

But the
brain “fills
in the gaps”
with familiar
figures



Kanizsa triangle

Deep Learning for visual recognition



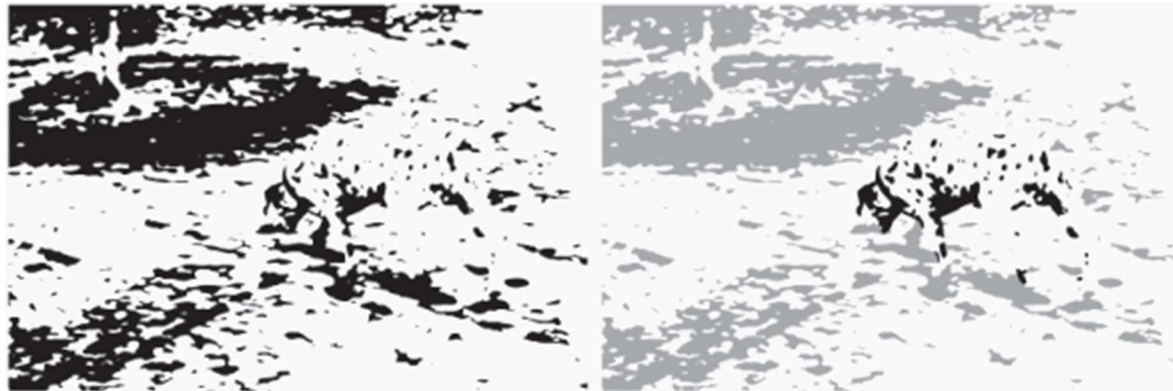
What should the system recognize?

Deep Learning for visual recognition



What should the system recognize?

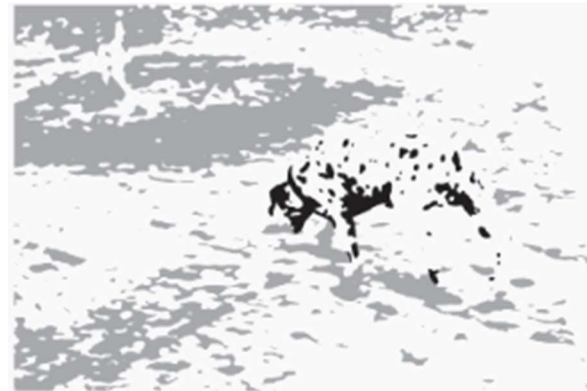
Deep Learning for visual recognition



Deep Learning for visual recognition



Yet, black dots are
just unconnected
data points.
Conceptual
interpretation makes
it a Dalmatian dog



Input: perception

☐ Visual

- Image
 - Pattern recognition: closure

1; 2; 3; 4; 5; _; 7; 8; 9;

Data imputation. What should be in the empty place?

Input: perception

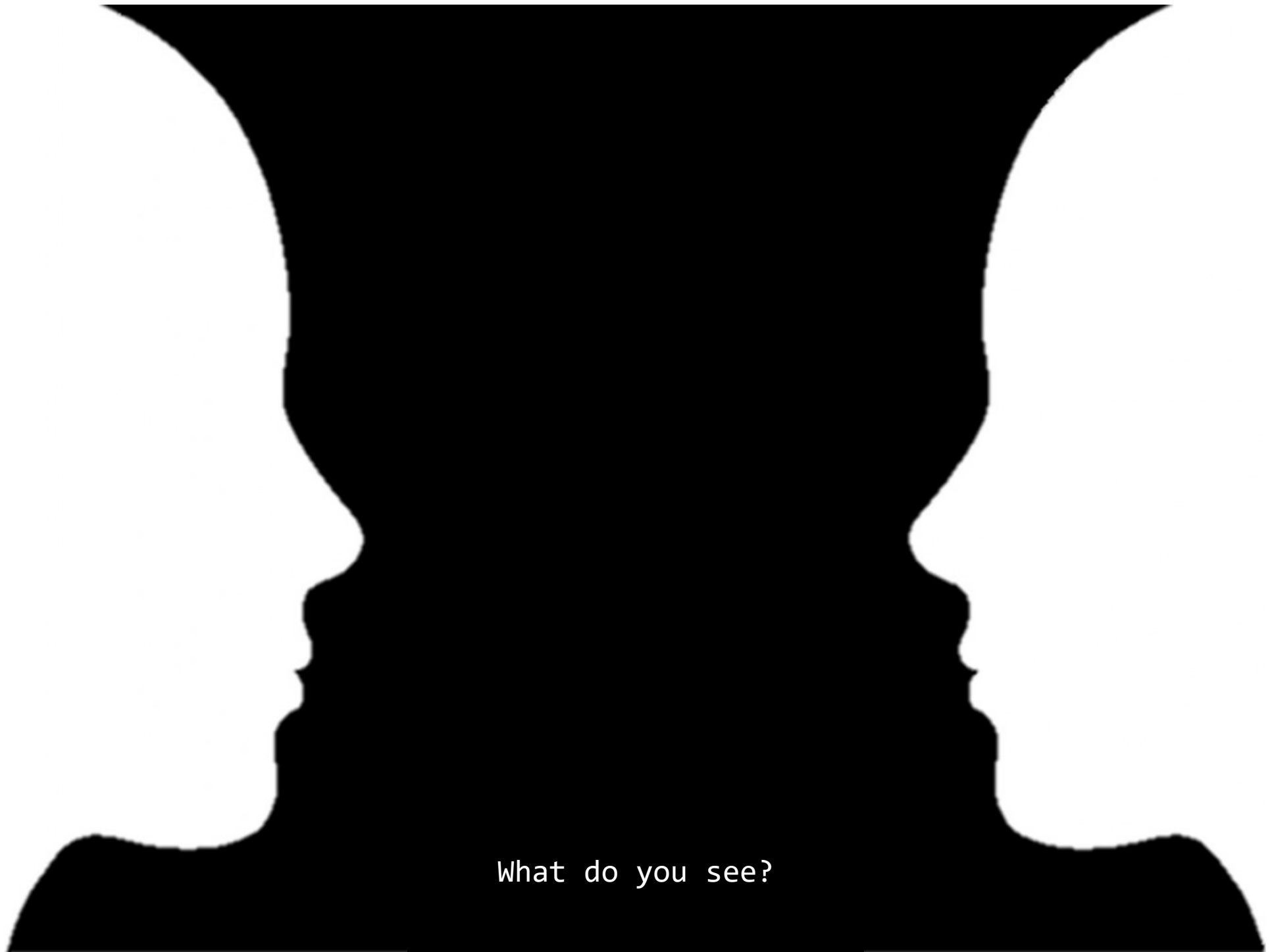
☐ Visual

- Image
 - Pattern recognition: closure

1; 2; 3; 4; 5; _; 7; 8; 9;

Data imputation. What should be in the empty place?

The most rational answer is: Don't know



What do you see?



Now, what do you see?
(as well)

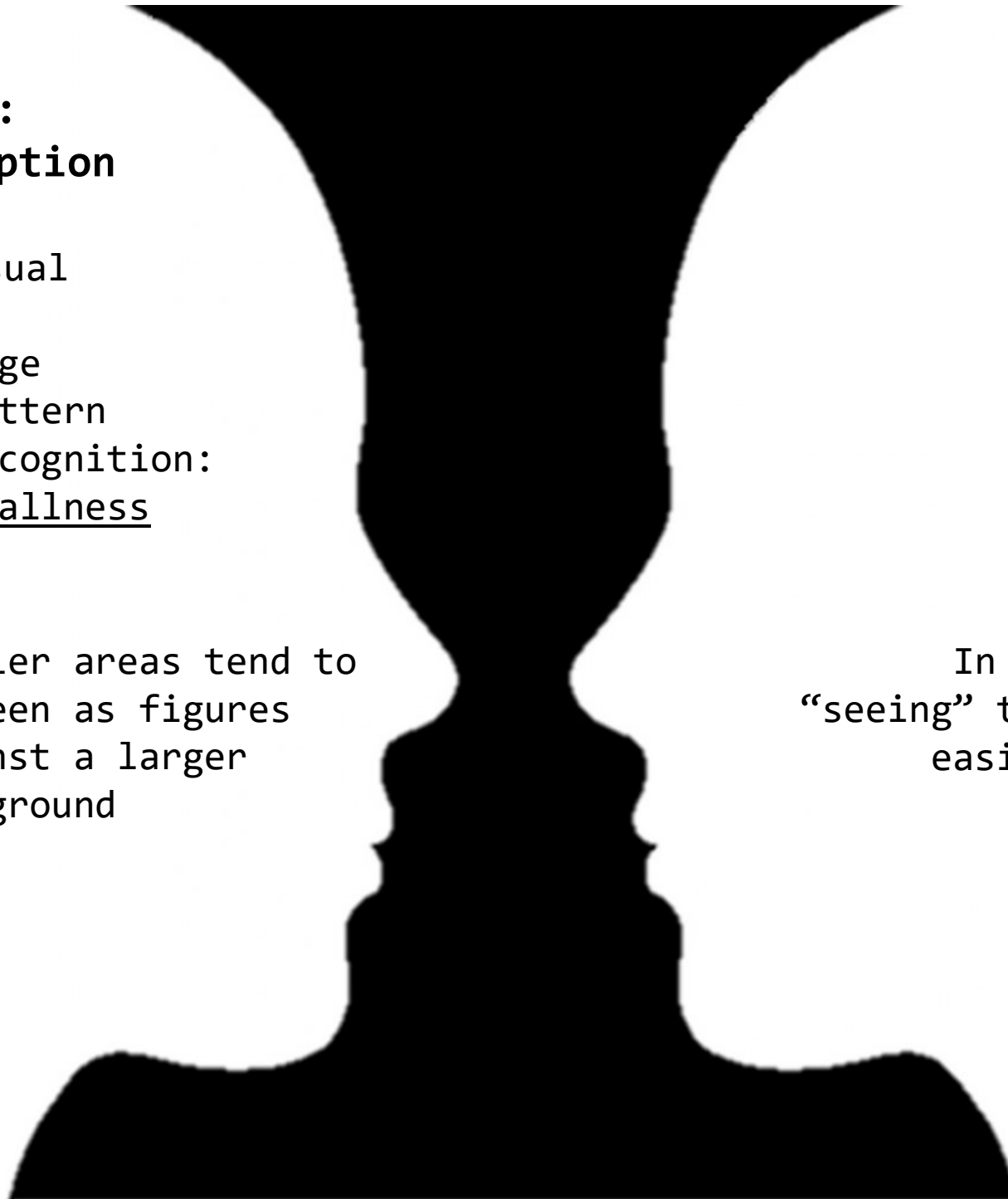
Input: perception

□ Visual

- Image
 - Pattern recognition:
smallness

Smaller areas tend to be seen as figures against a larger background

In the image, “seeing” the vase is easier when it is smaller





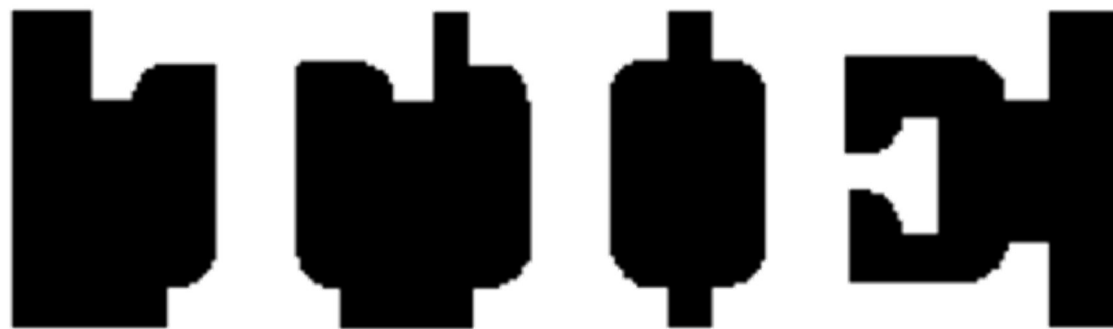
What are these?



What are these?



What are these?



What are these?
They are letters



The letters spell the word TIE

Input: perception

□ Visual

- Image
 - Pattern recognition: surroundedness

Areas which are surrounded by others tend to be seen
as independent figures



At first, most people will not see the word “TIE” in white letters
against a black background



Input: perception

□ Visual

- Image
 - Pattern recognition

Consequences

Therefore, pattern recognition is **NO** property of the data
(counting frequencies may do just as well)

To turn data into information, the following happens:

DATA —————→ **INFORMATION**

Input: perception

□ Visual

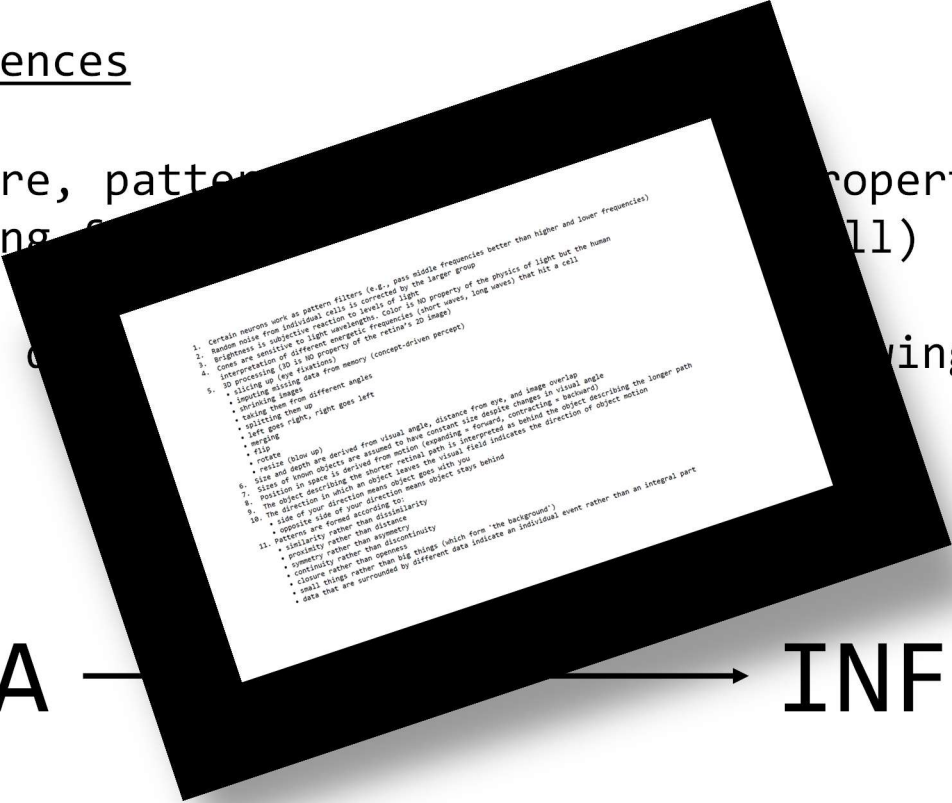
- Image
 - Pattern recognition

Consequences

Therefore, pattern recognition is a property of the data (counting on the brain)

To turn data into information, something happens:

DATA → INFORMATION

- 
1. Certain neurons work as pattern filters (e.g., pass middle frequencies better than higher and lower frequencies)
 2. Auditory noise from individual cells is corrected by the larger group
 3. Brightness is subjective reaction to levels of light
 4. Colors are sensitive to light wavelengths. Color is no property of the physics of light but the human interpretation of different energetic frequencies (short waves, long waves) that hit a cell
 5. 3D processing (3D is no property of the retina's 2D image)
 - slicing up eye fixations
 - stitching missing data from memory (concept-driven percept)
 - stitching images
 - taking them from different angles
 - splitting them up
 - left eye-right, right goes left
 - merging
 - flip
 - rotate
 - resize (blow up)
 6. Size and depth are derived from visual angle, distance from eye, and image overlap
 7. Size of human objects are assumed to have constant size despite changes in visual angle
 8. Position in space is derived from motion (expanding = forward, contracting = backward)
 9. Position describing the shorter retinal path is interpreted as behind the object describing the longer path
 10. The direction in which an object goes with you
 - the direction of your direction means object stays behind
 - side of your direction means object stays behind
 11. Patterns are formed according to:
 - opposite size of your direction
 - similarity rather than distance
 - proximity rather than asymmetry
 - symmetry rather than openness
 - continuity rather than discontinuity
 - closure rather than big things (which form 'the background')
 - small things rather than big things (which form 'the background')
 - data that are surrounded by different data indicate an individual event rather than an integral part

DATA

1. Certain neurons work as pattern filters (e.g., pass middle frequencies better than higher and lower frequencies)
2. Random noise from individual cells is corrected by the larger group
3. Brightness is subjective reaction to levels of light
4. Cones are sensitive to light wavelengths. Color is NO property of the physics of light but the human interpretation of different energetic frequencies (short waves, long waves) that hit a cell
5. 3D processing. 3D is NO property of the retina's 2D image
 - slicing up (eye fixations)
 - imputing missing data from memory (concept-driven percept)
 - shrinking images
 - taking them from different angles
 - splitting them up
 - left goes right, right goes left
 - merging
 - flip
 - rotate
 - resize (blow up)
6. Size and depth are derived from visual angle, distance from eye, and image overlap
7. Sizes of known objects are assumed to have constant size despite changes in visual angle
8. Position in space is derived from motion (expanding = forward, contracting = backward)
9. The object causing the shorter retinal path is interpreted as behind the object causing the longer path
10. The direction in which an object leaves the visual field indicates the direction of object motion
 - the side of your direction means object goes with you
 - opposite side of your direction means object stays behind
11. Patterns are formed according to:
 - similarity rather than dissimilarity
 - proximity rather than distance
 - symmetry rather than asymmetry
 - continuity rather than discontinuity
 - closure rather than openness
 - small things rather than big things (which form 'the background')
 - data that are surrounded by different data indicate an individual event rather than an integral part

INFORMATION

Virtual Tutor Summary



We casually speak about data as if they are lying around like rocks and sand and we habitually 'extract' information from the data as if chemical content (polyphenol) seeps out of infused green tea leaves

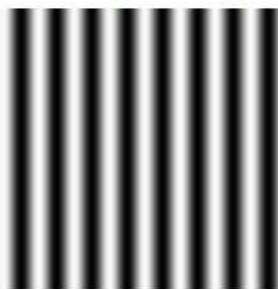
However, information is not neutral. The eye filters, selects, and changes data so to make an interpretation of the world about us. Color is not a property of light but of the eye's receptors. 3D vision results from manipulations by the brain to create a depth illusion. The brain uses heuristics such as similarity, continuity, and proximity to make sense of information that passes the eye. Patterns are not the result of recognition but rather of human construction

ALL computer analysis depends on human analysis!!

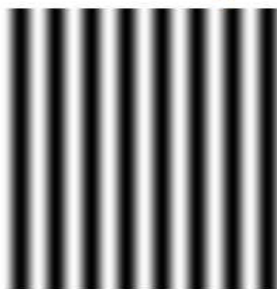
Visual Stimuli and Cell Responses

Learn about visual stimuli

Drifting Gratings



Static Gratings



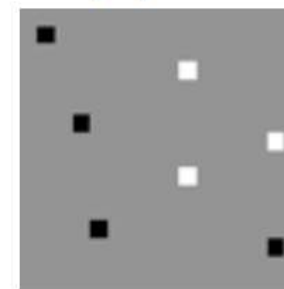
Natural Scenes



Natural Movies

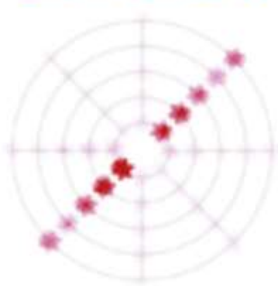


Locally Sparse Noise



Browse cell responses

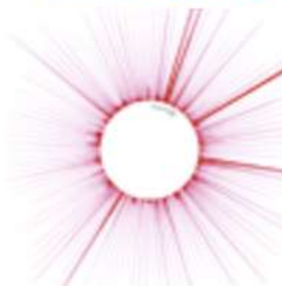
Drifting Gratings



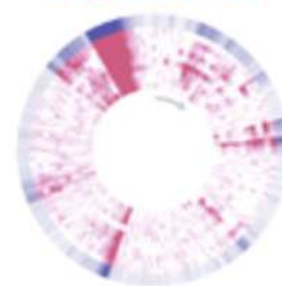
Static Gratings



Natural Scenes



Natural Movies



Locally Sparse Noise

